Minimal Decidable Site for the Madhyasth–Darshan Classifying Topos via Single-Flag Morleyisation

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Abstract

We present a *finite, machine–checkable* categorical semantics for Madhyasth Darshan by building a *decidable coherent site* on exactly **27** atomic predicates (26 geometric + one logical). A single, conservative Morleyisation adds the flag R_{walls} to name the otherwise undecidable Wall disjunction; all other families are already exclusive or nested, so no further flags are needed. Using Johnstone's Comparison Lemma we prove

$$\mathrm{Sh}\big(\mathcal{C}^{*}_{\mathrm{MD}}, \mathcal{J}^{*}\big) \;\simeq\; \mathscr{W}(T_{\mathrm{MD}}),$$

thereby preserving the intuitionistic internal logic while making equality decidable on the site. Five Lawvere–Tierney nuclei $\Box_J, \Box_D, \Box_b, \Box_r, \Box_\ell$ model the MD value layers and *preserve complements on ledger–clopens*, so decidability propagates through the entire modal stack. Geometrically, the ten-tier *World–Family* is realised as a clopen flag sheaf, and the chain $\mathrm{CK} \subset \mathrm{HCK} \subset \mathrm{SK}$ embeds as conservative open subtoposes, bridging cosmic, social, and personal viewpoints. The outcome is a minimal, upgrade-safe, topos-theoretic foundation that simultaneously aligns (i) intuitionistic logic, (ii) modal value layers, and (iii) MD's tiered ontology—opening the way to mechanised proofs and principled simulations.

Keywords: classifying topos; coherent site; Morleyisation; Lawvere–Tierney nucleus; modal transparency; Madhyasth Darshan; 10-Tier World-Family.

MSC2020: 18B25 (Toposes and related structures); 03G30 (Categorial logic, topoi); 03B45 (Modal logic).

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1 Introduction

Background. *Madhyasth Darshan* (MD) [18, 19] proposes a layered ontology ranging from personal conscience to cosmic order.[1] While MD has informed ethics and education, a *formal* mathematical universe capturing its axioms, value modalities, and tier structure has been missing. The natural candidate— the classifying topos $\mathscr{W}(T_{\mathrm{MD}})$ of the geometric theory T_{MD}— is infinitary and undecidable, hence unsuitable for mechanised proof or constructive computation (cf. categorical logic, classifying toposes, and the tripos pipeline [10, 5, 15]).

Main aim. We present a *finite, decidable, and modality-stable* site that *classifies the same topos*. The construction resolves three foundational questions: *(i)* **Minimal vocabulary:** how small can the primitive ledger be? *(ii)* Which logical extensions are *forced* by undecidability? *(iii)* Do the five MD value modalities preserve decidability?

Foundational setting. All reasoning takes place in the internal Heyting logic of a 1–topos $\mathcal{E}$. Our metaframework is ETCS; alternatively one may work in ZFC with a Grothendieck universe U, writing $\mathbf{Cat}_{\mathsf{U}}$ for U–small categories and taking sites/topoi to be U–small as needed. No classical axioms (LEM, Choice) are assumed unless explicitly indicated. Lawvere–Tierney topologies $j_X : \Omega \to \Omega$ determine open subtopoi $a_X \dashv i_X : \mathcal{E}_X \hookrightarrow \mathcal{E}$ and their idempotent, left-exact nuclei $\Box_X := i_X a_X$ (monads on $\mathcal{E}$; the induced comonad lives on $\mathcal{E}_X$) [4]. *Constructive alternatives.* Our stance is compatible with the programme of Algebraic Set Theory (AST) and constructive set theories interpreted in toposes; see Joyal–Moerdijk's AST and its predicative developments by Moerdijk–Palmgren [12, 3]. We remain in the ETCS/ZFC + Grothendieck framework for definiteness, but none of our core arguments depend on classical principles. Any use of Booleanity is either (i) *scoped to ledger–clopens* (finite Boolean algebra generated by the 26+1 atoms), or (ii) *confined to the resolved Stage–D subtopos* defined by $\neg R_{\mathrm{walls}}$, whose Boolean internal logic is established in the companion Paper A [23]. The results proved here do not rely on (ii).

Brahma and the SSS principle. We introduce the axiom **A0:** $\top \vdash \mathbf{1}_B$, which *establishes the omnipresent substratum* as the terminal object representing Brahma ($\mathbf{1}_B$ is fixed series-wide; see Paper 0 [22]). Every object X carries the unique map $X \to \mathbf{1}_B$. In MD this omnipresence is summarised by the *Soaked–Surrounded–Submerged (SSS)* dictum. Formally we use SSS in a *family-scoped* way: within each family (Walls Σ, Virtues V, Responsibilities R, Scales Sc) the atoms are exclusive; the Shells Δ form a *nested chain*; and *cross-family co-holding is permitted.* We keep this unfolded as geometric axioms A1–A9; the full SSS construction establishing orthogonality, nesting, and non-redundancy is proved in [20] (with the SU(5) representation-theoretic split in [21]).

Key results.

R1. Exact 26+1 ledger (minimal, non-redundant). The MD signature splits into 26 *geometric* atoms (Walls, Virtues, Responsibilities, Scales, Shells) plus a *single* logical flag R_{walls} introduced by Morleyisation to name the undecidable Wall disjunction; no other flags are required (Sections 2–5). The *exact* "26" are forced by the SSS projector construction;

[1] Foundational text: A. Nagraj, *Manav Vyavhār Darshan* [18]; consolidated modern edition: [19].

removing *any one* atom breaks the join-cover (non-vanishing of mixed projectors, [20]). (Provenance and roles are summarised in Appendix D.)

R2. Topos equivalence. By the Comparison Lemma, the sheaf topos $\mathrm{Sh}(\mathcal{C}^*_{\mathrm{MD}}, J^*)$ on this 27-generator site is *equivalent* to $\mathscr{W}(T_{\mathrm{MD}})$ (Section 7; see [1, Elephant A4] and the general classifying-topos pipeline [10, 5, 15]).

R3. Modal transparency (ledger–clopens). Each value nucleus $\Box_J, \Box_D, \Box_b, \Box_r, \Box_\ell$ preserves clopens generated by the ledger (finite meets and additivity on disjoint ledger unions), so equality remains decidable at *every* value layer (Section 8; monadic phrasing in [23]).

R4. 10-Tier World–Family *reinforced*: catalogue, equivalence, factorisation, elections. The ten MD "world tiers" assemble into a clopen flag sheaf forming a finite rooted *tree* on exactly ten nodes, with a bottom-up *election rule* determining parent links. Beyond earlier drafts, we now provide: (a) a concrete *catalogue* of Tier formulæ and core roles (Appendix E); (b) an *equivalence* between the older Δ–scheme and the canonical R_4/Scale scheme (Appendix F); (c) a *Steward factorization* theorem ensuring guidance lands in the guided stewards (Appendix G); and (d) an *instance opfibration* over levels formalising nomination/election as opcartesian lifts (Appendix H). The chain CK $\subset$ HCK $\subset$ SK embeds as conservative open subtoposes, linking cosmic, social, and personal perspectives (Section 9).

Logical stance and methodology. We work in the internal *intuitionistic* logic of a 1–topos and avoid Booleanisation (which would trivialise value modalities). Decidability is restored by a *single* Morley flag. Proofs are *geometric* and slice-stable; equalities of subobjects are stated *up to j-closure* when appropriate. For nuclei we use the reflector/inclusion adjunction $a_X \dashv i_X$ and the idempotent, left-exact monad $T_X = i_X a_X$ on the ambient topos (the induced comonad lives on the j_X-sheaf subtopos) [4]. Morleyisation is used conservatively in the sense of model theory (see, e.g., [11, 10]). A stronger, context-specific result—Boolean internal logic for the resolved (Stage–D) subtopos defined by $\neg R_{\mathrm{walls}}$—is established in the companion work [23]; we do not rely on that fact here.

Relation to other work. Section 10 contrasts our approach with Booleanisations, hyperconnected–localic factorisations, and higher-topos attempts, explaining why those either inflate the generator count or erase MD modalities. For background on factorisations/local maps see [7, 8]; for higher toposes see [13]. Technical ingredients (family–layer grid, Schur projectors, capacity bounds) are developed in [20, 21]; monadic and stability facts in [23].

Scope/Upgrade. All results here live in a 1–topos $\mathcal{E}$. The SSS constraints act internally on $\Omega_{\mathcal{E}}$ to govern *co-habitation* across slices (Def. 1.1). Our ontological anchors are fixed across the series: Brahma $\mathbf{1}_B$ (terminal of the Balance shell; Paper 0 [22]), the *Ātmā–atom* J as the fundamental object in SK, the *material atom* m in CK, and the *human-family* viewed as an *atom* in HCK. A globally coherent calculus of *relations-between-relations* (our *co-existence*, Def. 1.2) is deferred to a cohesive/$(\infty, 1)$–topos sequel; here we work 1-categorically and, where relevant, state results up to j-closure so they upgrade verbatim. Lawvere–Tierney nuclei are taken as idempotent, left-exact monads $i_X a_X$ [4]; dual Shape/Flat phenomena and the SK side's constitutional completeness are developed in the companions (Paper A for $\int \dashv \flat$ and shell commutation; Paper B for shape-level dualities and SK completeness) [23, 24]. After Stage–D the Ātmā–atom is *complete even without* a body—J no longer depends on any host, but *supports* the host (Paper 0) [22].

Definition 1.1 (Co-habitation)**.** Let $J \in \mathrm{SK}$ be an Ātmā–atom and $M \in \mathrm{CK}$ a material medium (e.g. an $A3$ body). With $a_8 \dashv i_8 : \mathrm{HCK} \hookrightarrow \mathcal{E}$ the open embedding at Tier 8, define the *host* of (J, M) by

$$\mathrm{Host}(J, M) \ := \ a_8(J \times M) \ \in \mathrm{HCK},$$

with unit (attachment) $\eta_{J,M} : J \times M \to i_8\mathrm{Host}(J, M)$. A *human-family atom* is any atomic object $F \in \mathrm{HCK}$ with a mono $F \hookrightarrow \mathrm{Host}(J, M)$.

Special statuses.

- **Free particle (FP) in the undivided monad.** Internally, J is anchored in $\mathbf{1}_B$. Write $\mathrm{FP}_{\mathrm{CK}}(J)$ for "J is *devoid of material atom in* CK," i.e. there is no $A1$ medium $m \hookrightarrow M$ with nontrivial attachment. In HCK, $\mathrm{Deluded}(J, M)$ abbreviates "$\mathrm{Host}(J, M)$ exists but $\mathrm{B}(J)$ fails" (a *deluded human*).
- **Persistence in** SK**.** Particles never lose their host Ātmā–atom in SK; the CK-reduct of J matches its material appearance, while SK endows higher properties via the $\int \dashv \flat$ dualities (Paper A, Paper B) [23, 24].
- **Isolated Ātmā–atom in** CK**.** The CK-visible trace of an isolated J cannot progress through B $\Rightarrow$ C $\Rightarrow$ proto-D $\Rightarrow$ D until it *enters* HCK via some medium M. After Stage–D, J is internally complete and does not *depend* on a host, but *gives support* to any host it may attach to (Paper 0) [22].

Definition 1.2 (Co-existence)**.** Given two co-habitations $H := \mathrm{Host}(J, M)$ and $H' := \mathrm{Host}(J', M')$ in HCK, a *co-existence datum* is a ledger–clopen relation $R \hookrightarrow H \times H'$ such that:

(i) **Value-invariance:** R is preserved by each value nucleus $\Box_X$.

(ii) **Projection coherence:** the projections $R \to H, H'$ are jointly conservative and commute with a_8, i_8 (hence with j-closure).

(iii) **Brahma anchoring:** R has a global element after composing to $\mathbf{1}_B$ (non-emptiness internally to the Balance shell).

Write $\mathrm{CoEx}(H, H')$ for the Heyting algebra of such relations (ordered by inclusion). *Upgrade note.* In the cohesive/$(\infty, 1)$ sequel we upgrade CoEx to a category of *relations-between-relations* (spans), compatibly with $\int \dashv \flat$ and shell commutation [23] and with standard duality frameworks [1, 13].

Modularity. This paper is self-contained: all definitions, axioms (A0–A9), the Morleyisation with its single flag, the 27-generator site, the topos equivalence, modal transparency on ledger-clopens, and the SK–HCK–CK bridge are presented and proved here. The core logical arguments—coherent site construction, topos equivalence, and modal transparency—are proven from the stated axioms. Citations to companion papers [20, 21, 22, 23] provide the deeper philosophical provenance and additional mathematical realisations, but are not necessary for verifying the deductive steps in this manuscript.

Organisation. Section 2 fixes the signature and gives early definitions (ledger–clopen; SK/HCK/CK as open subtopoi at tiers; stage taxonomy $A1, A2, A3, \mathrm{B}, \mathrm{C}, \text{proto-}D, \mathrm{D}$). Section 3 prints the coherent axioms (A0–A9) and recalls the family-scoped SSS. Section 5 performs the Morleyisation and proves decidability. Section 6 builds the 27-generator site. Section 7 establishes topos equivalence. Section 8 proves modal transparency on ledger-clopens. Section 9 realises the 10-Tier World–Family (as a finite rooted tree with bottom-up election) and the CK $\subset$ HCK $\subset$ SK bridge. Section 10 surveys alternatives. Section 11 outlines future work. Appendix A gives a two-map Boolean proof on the 26 geometric atoms (analysis $F : \Omega_{\mathcal{A}} \to \mathbf{2}^{\mathcal{A}}$; synthesis $G : \mathbf{2}^{\mathcal{A}} \to \Omega_{\mathcal{A}}$), formalising the analytic–synthetic duality. Appendix D maps the 26+1 symbols; the full SSS construction is cited in [20].

2 Signature and Glossary

We freeze the vocabulary for the remainder of the paper. All variables range over a single sort P.

> **Two-theory convention (applies to §§2–6).** The vocabulary below belongs to the *original* geometric theory T_{MD}. Wall exclusivity is *deliberately absent* here; it re-enters only in the Morleyised, decidable theory T^{*}_{MD} as Axiom A2*. See §3 for the axioms, §4 for the diagnosis, and §5 for the fix.

2.1 Sort

Symbol	Reading
P	world–point (ākāśic locus)

2.2 Geometric atoms (26)

Family	Primitive predicates	#
Walls	$\Sigma_1, \ldots, \Sigma_6$	6
Virtues	$V_1, \ldots, V_6$	6
Responsibilities	$R_1, \ldots, R_4$	4
Scales	$\mathrm{Sc}_1, \ldots, \mathrm{Sc}_5$	5
Shells	$\Delta_1, \ldots, \Delta_5$	5
Total geometric atoms		26

Definition 2.1 (Ledger–clopen)**.** A subobject $U \hookrightarrow X$ is a *ledger–clopen* iff, internally to $\mathcal{E}$, U is a finite disjoint union of atomic predicates drawn from the 26 geometric families above together with, where present, the single logical flag R_{walls} (Def. 2.3). Equivalently: U lies in the finite Boolean algebra generated by these 26+1 atoms under finite unions, finite meets, and complement in X.

Family scope and co-holding. Within each of V, R, Sc the atoms are *pairwise exclusive*. The shells Δ form a *nested chain*: $\Delta_{m+1} \Rightarrow \Delta_m$ for $m = 1, \ldots, 4$. *Cross-family co-holding is permitted:* conjunctions such as $\mathrm{Sc}_\ell \wedge V_j \wedge R_k \wedge \Sigma_i \wedge \Delta_m$ are consistent and used throughout. (Exclusivity for Walls Σ is intentionally *absent* in T_{MD} and reinstated as A2* only in T^{*}_{MD}; see §5.)

Provenance of the exact 26. The "family–layer grid" arises by fusing five value-layer Schur projectors with central family idempotents (companion [20]; SU(5) split summarised in [21]): five projectors P_k with block sizes $1, 2, 8, 18, 32$ (Shells, Scales, Responsibilities, Virtues, Walls) commute with family idempotents $e_{F,i}$ (one per atom in a family). The mixed operators $P_k e_{F,i}$ yield exactly 26 non-vanishing nuclei arranged as in the grid; their joins cover truth. This gives a *structural* (non ad hoc) origin for 6+6+4+5+5.

Lemma 2.2 (Non-redundancy of the 26 geometric nuclei)**.** *Let $\{\pi_\alpha\}_{\alpha<26}$ be the mixed nuclei $\pi_\alpha := P_{k(\alpha)}\, e_{F(\alpha),i(\alpha)}$. Then $\bigvee_{\alpha<26} \pi_\alpha(\top) = \top$, but for every β one has $\bigvee_{\alpha\neq\beta} \pi_\alpha(\top) < \top$.*

Sketch. Using $\sum_k P_k = 1$, $\sum_i e_{F,i} = 1$, and $P_k e_{F,i} = e_{F,i} P_k$,

$$\bigvee_{\alpha<26} \pi_\alpha(\top) = \bigvee_{k,i} P_k e_{F,i}(\top) = \bigvee_k P_k\Big(\bigvee_i e_{F,i}(\top)\Big) = \top.$$

The non-vanishing of each mixed projector $P_k e_{F,i}$ is proved in [20]. Removing π_β from the join subtracts a *non-zero idempotent* from $1 = \bigvee_{k,i} P_k e_{F,i}$, hence the residual join is $< \top$. □

Remark 2.3 (Wall exclusivity deferred)*.* The Wall cover $\Sigma_1 \vee \cdots \vee \Sigma_6$ is undecidable in T_{MD} and is *named* by a single Morley flag (next subsection). Exclusivity for Walls re-enters only in the decidable theory T^*_{MD} as Axiom A2* (§5).

2.3 Logical atom (Morley flag)

$$\boxed{R_{\text{walls}} : \ P \to \Omega \qquad R_{\text{walls}} \ \dashv\vdash \ \Sigma_1 \vee \cdots \vee \Sigma_6}$$

No other Morley symbols are required; the ledger thus contains $26 + 1 = 27$ atomic predicates. (§5 proves decidability and explains why a single flag suffices; see also the conservativity remark in §3.)

2.4 Value–layer nuclei

Five Lawvere–Tierney nuclei model the MD value layers. Writing $a_X \dashv i_X$ for the open reflector/inclusion, we use $T_X := i_X a_X$ as the idempotent, left-exact *monad* on the ambient topos; dually $a_X i_X$ is a (trivial) comonad on the j_X–sheaf subtopos. We denote by $\Box_X$ the induced closure on $\mathrm{Sub}(-)$ classified by $j_X : \Omega \to \Omega$.

Symbol	MD layer	Reading
$\Box_J$	Ātmā	depth (shell/permeative)
$\Box_D$	Buddhi	scope (scale/transparency)
$\Box_b$	Chitta	ought (resolution/omniscience)
$\Box_r$	Vṛtti	disposition (responsiveness)
$\Box_\ell$	Mana	reactivity (self-reliance)

Notation rule. Only these five symbols appear in theorem statements; any auxiliary nuclei used inside proofs are decorated (e.g. $\Box_{vr}$) and *defined* from the primaries.

Lemma 2.4 (Classifier transport for $\Box$-images)**.** *If $\chi_U : Y \to \Omega$ classifies $U \hookrightarrow Y$, then $j_X \circ \chi_U$ classifies $\Box_Y U$.*

Proof. Standard for Lawvere–Tierney topologies: the subobject classified by $j \circ \chi$ is the j-closure of U (see [4, §III.2]). Here we apply this for each value nucleus. □

2.5 Stages and evolution layers

We axiomatically name the material stages in CK and the Ātmā-internal stages in SK, and record the two evolution layers we use later.

Definition 2.5 (Material stages in CK)**.**

- $A1$ **(material atom).** A basic material atom $m \in \mathrm{CK}$ whose neighbourhoods are *microlinear* and whose incidence diagram along ledger–clopens has finite limits.
- $A2$ **(plant cell).** A finite connected colimit of $A1$–atoms equipped with a coherence map for nutrient/flow predicates; this fixes "plant cell" as a designated $A2$.
- $A3$ **(body/brain medium).** A stratified finite diagram of $A2$–objects with typed organ maps and a global "body" predicate; used as the medium in $\mathrm{Host}(J, M)$.

Definition 2.6 (Ātmā-internal stages in SK)**.** For an Ātmā–atom $J \in \mathrm{SK}$, we use monotone internal flags $\mathrm{B}(J) \Rightarrow \mathrm{C}(J) \Rightarrow \text{proto-D}(J) \Rightarrow \mathrm{D}(J)$, representing the successive completions (Paper 0 and Paper A). After Stage–D, J is complete even without a body (cf. §1.1).

Remark 2.7 (Two evolution layers). (1) $A1 \Rightarrow A2 \Rightarrow A3$ is *material composition* in CK. (2) $\mathrm{B} \Rightarrow \mathrm{C} \Rightarrow \text{proto-D} \Rightarrow \mathrm{D}$ is the *Ātmā-internal* evolution in SK. Realisation at host level occurs via $\mathrm{Host}(J, M) := a_8(J \times M) \in \mathrm{HCK}$ once an $A3$ medium M is present.

2.6 Ten–Tier World–Family (10TWF) and knowledge slices

We formalise the ten world–tier flags as clopen subterminals and organise them into a rooted tree.

Definition 2.8 (10TWF flags and election)**.** Let $W_k \hookrightarrow 1$ $(k = 1, \dots, 10)$ be ledger–clopen flags in $\mathcal{E}$ such that $\bigvee_{k=1}^{10} W_k = \top$ and $W_k \wedge W_{k'} = \bot$ for $k \neq k'$. An *election relation* $\mathrm{Elig}(j, k)$ on $\{2, \dots, 10\} \times \{1, \dots, 9\}$ selects for each $k \geq 2$ a unique parent $p(k)$ with $\mathrm{Elig}(p(k), k)$. The induced graph on $\{W_k\}$ is the *10TWF tree.* (Explicit Elig is given in §9.)

Proposition 2.9 (Tree property and exact cardinality)**.** *The 10TWF digraph from Def. 2.8 is a finite rooted tree on exactly ten nodes.*

Sketch. Disjointness and the cover give ten distinct nodes. Uniqueness of $p(k)$ for $k \geq 2$ forbids cycles and guarantees a unique incoming edge, hence a rooted tree with root W_1. $\square$

Definition 2.10 (Knowledge slices $\mathrm{SK}, \mathrm{HCK}, \mathrm{CK}$ as open subtopoi)**.** Let $a_t \dashv i_t : \mathcal{E}_t \hookrightarrow \mathcal{E}$ be the open subtopos determined by the tier nucleus generated by $\bigvee_{j \leq t} W_j$. Define

$$\mathrm{SK} := \mathcal{E}_7, \qquad \mathrm{HCK} := \mathcal{E}_8, \qquad \mathrm{CK} := \mathcal{E}_9,$$

so that $\mathrm{CK} \hookrightarrow \mathrm{HCK} \hookrightarrow \mathrm{SK}$ are conservative open inclusions (the $t = 10$ flag is terminal).

Remark 2.11 (Modal transparency on ledger–clopens). By Lemma 2.4, each value nucleus $\Box_J, \Box_D, \Box_b, \Box_r, \Box_\ell$ preserves complements on ledger–clopens. In particular, the 10TWF flags W_k and all tier unions are stable under value modalities (used in §8).

2.7 Derived shorthands and reading map

- *Complement:* $\overline{U}$ denotes the complement of a subobject $U \hookrightarrow X$.
- *Tier predicates:* legacy T_k may be read as W_k for $k = 0, \ldots, 9$.
- *Host notation:* $\mathrm{Host}(J, M) := a_8(J \times M) \in \mathrm{HCK}$ (Def. 1.1).

Need	**Where to look**
Full axiom list (A0–A9)	Section 3
Why only R_{walls} is added	Section 5
Decidable site construction	Section 6
Topos equivalence proof	Section 7
Modal complement preservation (ledger)	Section 8
10TWF & SK–HCK–CK bridge	Section 9

3 Coherent Axiom Set

We work over a single sort P (*world–points*). All family predicates $\Sigma_i, V_j, R_k, \mathrm{Sc}_\ell, \Delta_m$ are unary $P \to \Omega$. Geometric sequents are written $\varphi(x) \vdash \psi(x)$ or $\vdash \varphi(x)$ with the free variable $x{:}P$ understood. Throughout this section we use only coherent (in particular, finitary geometric) reasoning; standard background on coherent theories and classifying toposes may be found in [4, Chs. III–IV] and [5, Ch. 2].

$$\top \ \vdash \ \mathbf{1}_B \tag{A0}$$

Brahma inhabited; terminal object / universal container.

$$\vdash \ \Sigma_1(x) \ \vee \ \cdots \ \vee \ \Sigma_6(x) \tag{A1}$$

Wall cover.
A2 is *intentionally omitted* in T_{MD}. (*No* Wall exclusivity here; this is the sole undecidability trigger. It re-enters as **A2*** only in the Morleyised theory T^*_{MD}, see §5.)

$$\vdash \ V_1(x) \ \vee \ \cdots \ \vee \ V_6(x) \tag{A3}$$
$$V_i(x) \wedge V_j(x) \ \vdash \ \bot \qquad (1 \le i < j \le 6) \tag{A4}$$

Virtue cover and pairwise exclusivity (non-substitutability).

$$\vdash \ R_1(x) \ \vee \ \cdots \ \vee \ R_4(x) \tag{A5}$$
$$R_i(x) \wedge R_j(x) \ \vdash \ \bot \qquad (1 \le i < j \le 4) \tag{A6}$$

Responsibility cover and pairwise exclusivity.

$$\vdash \ \mathrm{Sc}_1(x) \ \vee \ \cdots \ \vee \ \mathrm{Sc}_5(x) \tag{A7}$$

$$\mathrm{Sc}_i(x) \wedge \mathrm{Sc}_j(x) \ \vdash \ \bot \qquad (1 \leq i < j \leq 5) \tag{A8}$$

Scale cover and pairwise exclusivity.

$$\Delta_{m+1}(x) \ \vdash \ \Delta_m(x) \qquad (m = 1, \dots, 4) \tag{A9}$$

Shell nesting $\Delta_{m+1} \Rightarrow \Delta_m$.

SSS schema (family-scoped, explicit). Covers (A1, A3, A5, A7); intra-family exclusivity (A4, A6, A8); and Shell nesting (A9). *No cross-family exclusion is assumed*: conjunctions such as $\mathrm{Sc}_\ell \wedge V_j \wedge R_k \wedge \Sigma_i \wedge \Delta_m$ are consistent.

Notes and forward references.

- **Coherence.** Axioms **A0–A9** are finitary geometric sequents; hence the base theory T_{MD} is coherent.
- **Undecidability locus.** The deliberate omission of Wall exclusivity (A2) is the *only* source of undecidable equality; all other families form disjoint covers.
- **Morleyisation (preview).** In §5 we pass to the decidable theory T^*_{MD} by **(i)** adding a Morley flag R_{walls} with biconditionals **A11a–b** ($R_{\mathrm{walls}}(x) \dashv\vdash \Sigma_1 \vee \cdots \vee \Sigma_6$), and **(ii)** reinstating Wall exclusivity as **A2***. This is a *definitional (conservative)* extension; proofs transport, and equality on ledger–clopens becomes decidable.
- **Modal layer (semantic, not axiomatic).** Value nuclei $\square_J, \square_D, \square_b, \square_r, \square_\ell$ are treated at the topos level later; they are *not* part of the syntactic theory T_{MD}.
- **Dependencies used later.** The 27-atom minimal-ledger and capacity results (§§6–7) cite the *Irreducibility Lemma*, *Sufficiency Lemma*, and the *Diagonal-split Lemma* (see Appendix B).

MD alignment (micro-note). A1/A3/A5/A7 record recognisable states at each value family; A4/A6/A8 enforce non-substitutability *within* a family; A9 encodes rank-wise refinement for Shells. The sole undecidability (A2 omitted) mirrors the "Wall" ambiguity that is resolved at Stage–D by asserting $\neg R_{\mathrm{walls}}$ in the companion analysis.

4 Diagnosing Undecidable Equality

Only one predicate family—the *Walls*—fails to yield a complemented diagonal in the original theory T_{MD}. All other families are already decidable in the relevant sense, so adding *exactly one* Morley flag suffices. [2]

Convention. For a family $F = \{F_i\}$ of unary predicates on P, write the *family–union* subobject as $R_F := \bigvee_i F_i \hookrightarrow P$. We call the equality on R_F *decidable* iff its diagonal $\Delta_{R_F} \hookrightarrow R_F \times R_F$ is complemented (internally in $\mathcal{E}$). For Shells we do *not* form a disjoint union (they are nested): Shell literals are used conjunctively with other atoms and do not trigger undecidability.

[2] Philosophically, the Walls encode the six agitations of outward-oriented Mana; see Appendix D.

4.1 Complemented vs. uncomplemented diagonals

Lemma 4.1 (Exclusive covers have rectangular complements)**.** *Let $\{F_i\}_{i=1}^n$ be a finite family of subobjects of P with $\vdash R_F := \bigvee_i F_i$ (cover) and $F_i \wedge F_j \vdash \bot$ for $i \neq j$ (exclusivity). Then in $P \times P$ one has*

$$\overline{\Delta_{R_F}} = \bigvee_{i \neq j} F_i \times F_j,$$

hence Δ_{R_F} is complemented and equality on R_F is decidable.

Proof idea. Disjointness gives $R_F \times R_F = \bigvee_{i,j}(F_i \times F_j)$ with pairwise disjoint pieces, and $\Delta_{R_F} = \bigvee_i (F_i \times F_i)$. The off–diagonal union $\bigvee_{i \neq j} F_i \times F_j$ is disjoint from Δ_{R_F} and their union is $R_F \times R_F$. □

Family	**Cover?**	**Exclusivity?**	**Complemented?**	**Verdict**
Walls Σ	yes (A1)	*no*	**No**	undecidable
Virtues V	yes (A3)	yes (A4)	Yes	decidable
Responsibilities R	yes (A5)	yes (A6)	Yes	decidable
Scales Sc	yes (A7)	yes (A8)	Yes	decidable
Shells Δ	chain (A9)	n/a	Yes†	decidable

† There is no disjoint family union for Shells; instead, Shell literals occur only in conjunction with other atoms. This usage does not create undecidable disjunctions in T_{MD}.

4.2 Step-1 ledger (what needs a flag?)

Family	**Complemented?**	**Morley flag needed?**
Walls Σ	No	R_{walls}
Virtues V	Yes	—
Responsibilities R	Yes	—
Scales Sc	Yes	—
Shells Δ	Yes	—

4.3 Minimality

Lemma 4.2 (Singleton-flag irreducibility)**.** *If no new predicate is added to the language, the Wall family–union diagonal Δ_{R_Σ} remains uncomplemented; hence at least one extra predicate is necessary.*

Proof idea. Without exclusivity, any geometric attempt to complement Δ_{R_Σ} uses a finite union of rectangles $\Sigma_i \times \Sigma_j$ with $i \neq j$. But overlaps $\Sigma_i \wedge \Sigma_j$ are consistent in T_{MD}, so each such rectangle meets the diagonal at equal pairs (x, x) where both $\Sigma_i(x)$ and $\Sigma_j(x)$ hold. Thus no geometric complement exists. Formal details are in Appendix B. □

Lemma 4.3 (Sufficiency)**.** *Adding the single flag R_{walls} (biconditionals A11a–b) and reinstating exclusivity (A2*) complements every previously undecidable Wall diagonal.*

Proof idea. With A2*, $\Sigma_1, \ldots, \Sigma_6$ are disjoint and Lemma 4.1 applies. The complement of $\Delta_{R_{\mathrm{walls}}}$ is the clopen union $\bigvee_{i \neq j} \Sigma_i \times \Sigma_j$, disjoint from the diagonal by A2*. See Appendix B. □

Theorem 4.4 (27-atom minimality)**.** *Exactly* 26 *geometric plus* 1 *logical* = 27 *generators are necessary* and *sufficient for decidable equality in the Madhyasth Darshan theory.*

Proof sketch. Necessity by Lemma 4.2. Sufficiency by Lemma 4.3 and the Diagonal–split Lemma 5.1 (full version in Appendix B). No other undecidable disjunctions arise. □

4.4 Morleyisation: definitional move and outcome

Proposition 4.5 (Conservativity / definitional extension)**.** *Let* T_{MD} *be given by A0–A9. Form* T^*_{MD} *by adjoining a new unary predicate* R_{walls} *with biconditionals* $R_{walls}(x) \dashv\vdash \Sigma_1(x) \vee \cdots \vee \Sigma_6(x)$ *(A11a–b) and reinstating Wall exclusivity (A2*). Then the forgetful functor* $\mathbf{Mod}(T^*_{\mathrm{MD}}) \to \mathbf{Mod}(T_{\mathrm{MD}})$ *is an equivalence; in particular, any sentence in the original language is provable from* T_{MD} *iff it is provable from* T^*_{MD}.

Proof sketch. Standard Morleyisation: adding a predicate naming an existing formula with $\dashv\vdash$ equivalences is definitional (hence conservative); see, e.g., [11, Ch. 3], [10, §I.5], [5, Ch. 3]. □

$$\boxed{R_{\mathrm{walls}} \;\dashv\vdash\; \Sigma_1 \vee \cdots \vee \Sigma_6 \quad \text{and} \quad \Sigma_i \wedge \Sigma_j \vdash \bot \; (i \neq j) \;\; \text{in } T^*_{\mathrm{MD}}.}$$

All further construction relies on this *sole* logical atom; the other families require no additional flags. On ledger–clopens, decidability propagates through value nuclei via classifier transport (Lemma 2.4).

5 Finite Morleyisation of AX−MD

The Wall cover

$$\top \;\vdash\; \Sigma_1 \vee \cdots \vee \Sigma_6 \tag{A1}$$

is the unique source of undecidable equality in T_{MD} (cf. §4). Morleyisation *names* this union, reinstates exclusivity, and replaces (A1) by an equivalent pair of coherent sequents.

5.1 Enlarged language

$$\Sigma^*_{\mathrm{MD}} \;=\; \Sigma_{\mathrm{MD}} \;\cup\; \{R_{\mathrm{walls}}(-)\},$$

where R_{walls} is a new unary predicate on P. The language now has 26 geometric + 1 logical = 27 atoms.

5.2 Syntactic rewrite

Delete (A1) and add the biconditionals naming the union:

$$R_{\mathrm{walls}}(x) \;\vdash\; \Sigma_1(x) \vee \cdots \vee \Sigma_6(x), \qquad \Sigma_1(x) \vee \cdots \vee \Sigma_6(x) \;\vdash\; R_{\mathrm{walls}}(x) \tag{A11a–b}$$

together with *Wall exclusivity*:

$$\Sigma_i(x) \wedge \Sigma_j(x) \;\vdash\; \bot \quad (i \neq j) \tag{A2*}$$

The resulting coherent theory is

$$T^*_{\mathrm{MD}} \;:=\; \mathrm{Morley(AX{-}MD)} \;=\; (T_{\mathrm{MD}} \setminus \{A1\}) \;\cup\; \{A11a, b,\; A2^*\}.$$

As shown in Proposition 4.5 (end of §4), this passage is a *definitional (conservative)* extension.

5.3 Decidability of the Wall flag

Lemma 5.1 (Diagonal–split template)**.** *If $R \cong \Sigma_1 \sqcup \cdots \sqcup \Sigma_n$ is a finite* disjoint *union in a coherent theory, then the diagonal $\Delta_R \hookrightarrow R \times R$ is complemented (clopen).*

Idea. Finite coproducts are disjoint and stable under pullback, so $R \times R \cong \left(\bigsqcup_i \Sigma_i \times \Sigma_i\right) \sqcup \left(\bigsqcup_{i \neq j} \Sigma_i \times \Sigma_j\right)$. The first summand is Δ_R; the second is its clopen complement. A full categorical proof is given in Appendix B. □

Application. (A11a–b) plus A2* yield $R_{\text{walls}} \cong \Sigma_1 \sqcup \cdots \sqcup \Sigma_6$; by Lemma 5.1 the diagonal $\Delta_{R_{\text{walls}}}$ is clopen.

Corollary 5.2 (Decidable equality on families and ledger–clopens)**.** *In T^*_{MD}, equality is decidable on each family–union object (R_{walls}, $\bigvee_i V_i$, $\bigvee_k R_k$, $\bigvee_\ell \mathrm{Sc}_\ell$); and more generally on every* ledger–clopen *$U \hookrightarrow X$ (Def. 2.1).*

Sketch. For V, R, Sc: covers+exclusivities (A3–A8) give disjoint unions, so apply Lemma 5.1. For ledger–clopens: by definition U is a finite *disjoint* union of atomic predicates (and possibly R_{walls}); the off–diagonal rectangle argument gives a clopen complement of Δ_U. □

Remark 5.3 (Propagation through value nuclei)*.* By classifier transport (Lemma 2.4 in §2), each value nucleus $\Box_J, \Box_D, \Box_b, \Box_r, \Box_\ell$ preserves complements on ledger–clopens, so decidability persists across value layers (used in §8).

5.4 Coherence and finiteness

All axioms remain finite geometric sequents; with 27 primitives the syntactic site is effective to present and its covering sieves are decidable.

5.5 Forward link to the finite site

Let

$$(\mathcal{C}^*_{\text{MD}}, J^*) \;:=\; \text{the coherent syntactic site of } T^*_{\text{MD}}.$$

In Section 6 we list the 27 generators explicitly (26 geometric + R_{walls}) and the covering families; we also verify the promised micro–fact that *shells contribute no sieve* (they are nested only). Section 7 then applies the Comparison Lemma to obtain $\mathrm{Sh}(\mathcal{C}^*_{\text{MD}}, J^*) \simeq \mathscr{W}(T_{\text{MD}})$.

6 The Decidable Coherent Site

Logical stance. All arguments in this section are intuitionistic (Heyting) and use only: finite limits, disjoint finite coproducts, regular images, and Lawvere–Tierney nuclei. Standard facts about generated Grothendieck topologies and sheaves follow [1, §C2]; see also [4, Chs. III–IV] and [5, Ch. 2].

6.1 Underlying preorder and generators

Let $\mathcal{C}^*_{\mathrm{MD}}$ be the coherent preorder of formulas of T^*_{MD} modulo provable equivalence, with an arrow $\varphi \to \psi$ when $T^*_{\mathrm{MD}} \vdash \varphi \Rightarrow \psi$. Its *indecomposable generators* are the 27 atomic literals of the ledger (Walls/Virtues/Responsibilities/Scales; Shells appear only as *conjuncts*). An explicit list is recorded in Appendix A. Finite conjunctions, finite disjunctions, and existential quantification are computed in the usual syntactic way.

6.2 Generating sieves and the topology J^*

We take the Grothendieck topology J^* to be *generated* by four families of covering sieves corresponding to the coherent covers of the primitive families; Shells contribute no generating sieve because they form a nested chain (A9).

Definition 6.1 (Generation of J^*)**.** J^* is the *smallest* Grothendieck topology on $\mathcal{C}^*_{\mathrm{MD}}$ that contains the sieves listed below and is stable under pullback along arrows of $\mathcal{C}^*_{\mathrm{MD}}$. Equivalently, J^* is the pullback-stable closure of these sieves in the sense of [1, §C2.2].

- **Virtues.** For the family $\{V_j\}_{j \in I_V}$, the coherent cover $\top \vdash \bigvee_j V_j$ yields the sieve $\{\, V_j \to \bigvee_j V_j \,\}_{j \in I_V}$ on $\bigvee_j V_j$.
- **Responsibilities.** Analogously, from $\top \vdash \bigvee_k R_k$ we take the sieve $\{\, R_k \to \bigvee_k R_k \,\}_{k \in I_R}$.
- **Scales.** From $\top \vdash \bigvee_\ell S_\ell$ we take $\{\, S_\ell \to \bigvee_\ell S_\ell \,\}_{\ell \in I_S}$.
- **Walls (Morley flag).** Using (A11a–b) and A2*, the cover $\top \vdash \Sigma_1 \vee \cdots \vee \Sigma_6$ induces on R_{walls} the sieve $\{\, \Sigma_i \to R_{\mathrm{walls}} \,\}_{i=1}^{6}$, which is jointly epimorphic (Appendix C, Joint epimorphy of generating sieves).

By construction J^* is pullback-stable; we collect the basic stability facts used later in Appendix C, Properties used in the density proof.

6.3 Coherence and subcanonicity

Proposition 6.2 (Coherence)**.** *$(\mathcal{C}^*_{\mathrm{MD}}, J^*)$ is a coherent site: covers are generated by* finite *families, and $\mathcal{C}^*_{\mathrm{MD}}$ has finite limits.*

Sketch. Finite limits exist in the syntactic preorder; generation uses only finitely many arrows per family (Virtues/Responsibilities/Scales/Walls). Details and a closure audit under pullback are in Appendix C, Properties used in the density proof. □

Proposition 6.3 (Subcanonicity)**.** *The topology J^* is subcanonical: every representable presheaf is a J^*-sheaf.*

Sketch. The covering criterion is given by coherent entailments, hence Yoneda sends joint epimorphic families to jointly covering families of matching families. See Appendix C, Subcanonicity, and [1, §C2.1]. □

6.4 Size audit and minimality

Proposition 6.4 (Tally of generators and covers)**.** *$\mathcal{C}^*_{\mathrm{MD}}$ has exactly 27 indecomposable generators. The topology J^* is generated by **four** families of finite sieves (Walls, Virtues, Responsibilities, Scales); Shells contribute none.*

Sketch. Direct inspection of the ledger atoms (Appendix A) gives the 27 indecomposables. Generating families are the four listed above; Shells are nested (A9), so no cover is required. The joint-epi and pullback-stability checks are in Appendix C, Joint epimorphy of generating sieves and Properties used in the density proof. □

Theorem 6.5 (Minimality of the Morleyised site)**.** *Removing any generator or any one of the four generating families of sieves destroys one of: decidability of the ledger, joint epimorphy of covers, or comparison-equivalence with $\mathscr{W}(T_{\mathrm{MD}})$.*

Idea. Walls require a single decidable name by the diagonal-split template; without (A11a–b) the Wall diagonal is undecidable. Adding flags to Virtues/ Responsibilities/Scales is redundant (their diagonals are already clopen), raising the ledger for no gain. See Appendix C for the diagnostic cases and Section 10.3 for the variant matrix. □

With $(\mathcal{C}^*_{\mathrm{MD}}, J^*)$ now fixed, Section 7 applies Johnstone's Comparison Lemma to obtain the topos-level equivalence with $\mathscr{W}(T_{\mathrm{MD}})$.

7 Topos Equivalence via the Comparison Lemma

We now show that the decidable coherent site constructed in Section 6 presents *exactly the same* classifying topos as the full syntactic site of the *original* theory T_{MD}:

$$\mathrm{Sh}(\mathcal{C}^*_{\mathrm{MD}}, J^*) \;\simeq\; \mathscr{W}(T_{\mathrm{MD}}).$$

By Johnstone's Comparison Lemma (e.g. [1, §C2.3]), a functor F between sites induces an equivalence of sheaf toposes if and only if F is *cover-preserving* and *dense*. We verify these in turn.

7.1 The comparison functor

Let

$$F:\ \mathrm{Syn}(T_{\mathrm{MD}}) \;\longrightarrow\; \mathcal{C}^*_{\mathrm{MD}}$$

be the *Morley translation* on objects and arrows.

On objects. For a formula φ in the language of T_{MD}, let φ^* be obtained by replacing each occurrence of the Wall disjunction $\Sigma_1 \vee \cdots \vee \Sigma_6$ by the single predicate R_{walls}. Set $F(\varphi) := \varphi^*$. Finite conjunctions, finite disjunctions, and existential quantification are preserved.

On arrows. If $T_{\mathrm{MD}} \vdash \varphi \Rightarrow \psi$ then, using (A11a–b), $T^*_{\mathrm{MD}} \vdash \varphi^* \Rightarrow \psi^*$; hence there is a unique entailment arrow $F(\varphi) \to F(\psi)$ in the preorder $\mathcal{C}^*_{\mathrm{MD}}$. Thus F is well-defined and functorial.

7.2 Cover preservation

It suffices to check the generating covers of the syntactic site. By the definition of $\mathcal{J}^*$ in §6.2—as the Grothendieck topology *generated* by the listed sieves and *stable under pullback*—F sends each generating cover to a $\mathcal{J}^*$-cover.

- **Virtues, Responsibilities, Scales.** Their cover-sequents are fixed by Morley translation, so F sends each to the corresponding $\mathcal{J}^*$-covering sieve listed in §6.2.
- **Walls.** The cover $\top \vdash \Sigma_1 \vee \cdots \vee \Sigma_6$ maps to the sieve $\{\ \Sigma_i \to R_{\text{walls}}\ \}_{i=1}^{6}$, which is $\mathcal{J}^*$-covering by (A11a) together with joint epimorphy from A2* (see Appendix C, Joint epimorphy of generating sieves).
- **Shells.** Shells form a nested chain (A9) and contribute no generating sieve; nothing to check.

7.3 Density of F

Lemma 7.1 (Local surjectivity on generators)**.** *Each indecomposable generator of $\mathcal{C}^*_{\text{MD}}$ is $\mathcal{J}^*$-covered by arrows in the image of F.*

Idea. For R_{walls} take the source object $\Sigma_1 \vee \cdots \vee \Sigma_6$; its inclusions $\Sigma_i \to \Sigma_1 \vee \cdots \vee \Sigma_6$ map under F to the covering sieve $\{\Sigma_i \to R_{\text{walls}}\}$. For any other literal ψ, F acts trivially and the sieve generated by id_ψ is covering. □

Proposition 7.2 (Density)**.** *For every $\varphi \in \mathcal{C}^*_{\text{MD}}$, the sieve generated by morphisms $F(\theta) \to \varphi$ is $\mathcal{J}^*$-covering.*

Idea. Objects of $\mathcal{C}^*_{\text{MD}}$ are obtained from the 27 generators by finite limits, finite coproducts, and images (Appendix C, Coherent envelope preserves generators). The class of objects covered by F-images is closed under these operations: covers are stable under pullback (Properties used in the density proof, Lemma C.4); finite sums of covering sieves cover sums (Lemma C.5); products are covered after pulling back along projections (Lemma C.6); and images preserve joint epimorphy (Lemma C.7). Since the generators are covered (Lemma 7.1), every object is covered. Full details are recorded in Appendix C, Properties used in the density proof. □

7.4 Equivalence of sheaf toposes

Since F is cover-preserving (Section 7.2) and dense (Proposition 7.2), the Comparison Lemma yields

$$\mathrm{Sh}(\mathcal{C}^*_{\text{MD}}, \mathcal{J}^*) \ \simeq\ \mathscr{W}(T_{\text{MD}}).$$

7.5 Modal nuclei transport

The five Lawvere–Tierney topologies are topos invariants; they induce *idempotent, left-exact monads* $T_X = i_X a_X$ on the ambient topos and dually comonads $a_X i_X$ on the j_X-sheaf subtopos. Under the above equivalence, subobject classifiers and these nuclei correspond along the induced isomorphism, so the Modal Transparency theorem (§8) holds identically in both presentations.

7.6 Foundational summary

- **Minimal decidable site.** 27 atomic generators, four finite generating cover sieves (Walls, Virtues, Responsibilities, Scales), clopen diagonals; Shells contribute no sieve.
- **Topos-level faithfulness.** $\mathrm{Sh}(\mathcal{C}^*_{\mathrm{MD}}, \mathcal{J}^*)$ is equivalent to the syntactic topos $\mathscr{W}(T_{\mathrm{MD}})$.
- **Modal stability.** The five value-layer nuclei act identically across the equivalence.

This completes the foundational construction. The next section establishes Modal Transparency in detail.

8 Modal Transparency—Value-Layer Nuclei Preserve Complements

By Section 7, the decidable site $(\mathcal{C}^*_{\mathrm{MD}}, \mathcal{J}^*)$ presents the syntactic topos of T_{MD}. We now show that each MD "value layer"—specified by a Lawvere–Tierney topology—*preserves complements* on the class we actually use: the *ledger–clopens* (finite disjoint unions of the 27 atomic predicates inside any object built from them; Def. 2.1).

Scope of Booleanity. All results here are proved in the ambient Heyting internal logic. In particular, Theorem 8.4 is scoped to *ledger–clopens*. By contrast, the *Stage–D* subtheory obtained by adding $\neg R_{\mathrm{walls}}$ yields a Boolean subtopos (companion [23]). Thus MDtopos remains intuitionistic overall, while the resolved (no–Wall) subuniverse has Boolean internal logic.

> **Modal Transparency (ledger–clopen form).** For every value layer $\Box \in \{\Box_J, \Box_D, \Box_b, \Box_r, \Box_\ell\}$ and every object X, if $U \hookrightarrow X$ is a ledger–clopen, then $\Box_X U$ is a ledger–clopen and $\overline{\Box_X U} = \Box_X \overline{U}$.

8.1 Lawvere–Tierney nuclei recalled

Each value layer is given by a Lawvere–Tierney map $j : \Omega \to \Omega$. Let $a : \mathcal{E} \to \mathcal{E}_j$ be the sheafification (reflector) and $i : \mathcal{E}_j \hookrightarrow \mathcal{E}$ the geometric inclusion. The induced *idempotent, left-exact monad* on $\mathcal{E}$ is

$$\Box \;:=\; i \circ a\,,$$

and for every object Y it acts as a closure operator $\Box_Y : \mathrm{Sub}(Y) \to \mathrm{Sub}(Y)$. If $U \hookrightarrow Y$ is classified by $\chi_U : Y \to \Omega$, then (by Lemma 2.4) $\Box_Y U$ is classified by $j \circ \chi_U$. We write $\Box \in \{\Box_J, \Box_D, \Box_b, \Box_r, \Box_\ell\}$ for the five fixed MD layers, and $\Box_X$ for its action on $\mathrm{Sub}(X)$.

8.2 Finite meets and additivity on disjoint ledger unions

Lemma 8.1 (Finite–meet preservation)**.** *For any value layer $\Box$ and subobjects $U, V \hookrightarrow X$, $\Box_X(U \cap V) = \Box_X U \cap \Box_X V$.*

Proof. $\Box = i \circ a$ is left exact; left exactness gives preservation of finite limits. $\square$

Lemma 8.2 (Finite additivity on disjoint ledger unions)**.** *Let $U_1, \dots, U_n \hookrightarrow X$ be* ledger–clopens *with $U_p \cap U_q = \varnothing$ for $p \neq q$. Then*

$$\Box_X\Big(\bigcup_{p=1}^{n} U_p\Big) \;=\; \bigcup_{p=1}^{n} \Box_X U_p.$$

Idea. In the ledger, a finite disjoint union is a finite coproduct $U_1 \sqcup \cdots \sqcup U_n$. Reflectors a preserve finite coproducts and i creates them, so $i \circ a$ preserves disjoint finite coproducts. Induct on n (the case $n = 2$ is the "binary union" case). □

Remark 8.3 (Scope). General nuclei need not preserve arbitrary joins. The lemma uses the disjointness guaranteed by (A2*, A4, A6, A8) to obtain additivity exactly on the ledger–clopen algebra.

8.3 Complement preservation

Theorem 8.4 (Modal Transparency). *If $U \hookrightarrow X$ is a ledger–clopen, then $\Box_X U$ is clopen and $\overline{\Box_X U} = \Box_X \overline{U}$.*

Proof. Write $X = U \cup \overline{U}$ with $U \cap \overline{U} = \varnothing$. By Lemma 8.1, $\Box_X(U \cap \overline{U}) = \Box_X U \cap \Box_X \overline{U} = \varnothing$. By Lemma 8.2, $\Box_X X = \Box_X(U \cup \overline{U}) = \Box_X U \cup \Box_X \overline{U}$. Hence $\Box_X U$ and $\Box_X \overline{U}$ are complements in $\mathrm{Sub}(X)$. □

8.4 Consequences and examples

Corollary 8.5 (Iterated transparency). *Any finite composite of the five nuclei preserves ledger–clopens; hence equality remains decidable throughout the modal stack.*

Example 8.6. Let

$$P(x) \;:=\; \mathrm{Sc}_1(x) \;\wedge\; V_4(x) \;\wedge\; \neg R_{\mathrm{walls}}(x) \quad \text{("individual scale, Kindness, resolved (no–Wall)").}$$

P is a ledger–clopen. Then $(\Box_r \circ \Box_b)(P)$ is again clopen by Theorem 8.4; deciding P remains crisp after Chitta– then Vṛtti–closure, matching the MD claim that disciplined reflection does not introduce ambiguity.

Remark 8.7 (Transfinite joins of layers). If a transfinite composite is interpreted as the join in the complete lattice of Lawvere–Tierney topologies, classifier transport shows that joins preserve the ledger–clopen property; complement preservation persists on this scope.

8.5 String–diagram intuition

$$\begin{tikzcd}
U \arrow[r, hook] \arrow[d, hook] & \Box_X U \arrow[d, hook] \\
\overline{U} \arrow[r, hook] & \Box_X \overline{U}
\end{tikzcd}$$

Unit monotonicity ($U \le \Box_X U$, $\overline{U} \le \Box_X \overline{U}$) and Lemmas 8.1–8.2 ensure the square exhibits complements mapped to complements.

8.6 Wrap–up

- **Modal transparency.** Each value layer preserves complements on ledger–clopens.
- **Decidable hierarchy.** Equality stays decidable at every modal level reached by iterating the nuclei.

- **No new generators.** The 27-atom ledger suffices for the entire modal stack.

Section 9 now puts these results to work, realising the 10-Tier World–Family and proving the $\mathrm{CK} \subset \mathrm{HCK} \subset \mathrm{SK}$ bridge.

9 10–Tier World–Family and the SK–HCK–CK Bridge

We formalise the 10–Tier World–Family (10TWF) as *ledger–clopen* tier flags and record two complementary orders on the knowledge slices $\mathrm{CK}, \mathrm{HCK}, \mathrm{SK}$. This yields a precise notion of a *human* as an SK–unit attached to a CK–body living in HCK, clarifies the CK–driven administrative hierarchy versus the SK–driven cultural network, and pins down where "co-existence" (relations between relations) sits beyond the present 1–topos.

9.1 Two orders on knowledge slices

Quantitative containment (external scope). There are conservative open embeddings of toposes

$$\mathrm{CK} \ \subset \ \mathrm{HCK} \ \subset \ \mathrm{SK},$$

read as "what the slice ranges over" (cosmic $\to$ conduct $\to$ self).

Qualitative/epistemic priority (internal adequacy). There is also a preorder of explanatory adequacy

$$\mathrm{SK} \ \succcurlyeq \ \mathrm{HCK} \ \succcurlyeq \ \mathrm{CK},$$

read as "how complete the inner constitution is." In Stage–D the Ātmā–atom's constitution is complete (full SSS alignment), furnishing humane conduct (HCK), which in turn organises the cosmic picture (CK). These orders are orthogonal; by §8, decidability on ledger–clopens is preserved across slices.

9.2 Human as SK+CK living in HCK

Definition 9.1 (Human object)**.** A *human* is a triple $\mathbf{H} = (J, B, \sigma)$ with (i) $J \in \mathrm{SK}$ an Ātmā–atom (five shells Δ; Stage–D is clopen), (ii) $B \in \mathrm{CK}$ a material body (e.g. an $A3$ body), and (iii) $\sigma \hookrightarrow \mathrm{Host}(J, B) \in \mathrm{HCK}$ an HCK–structure recording conduct/compatibility (virtue–responsibility constraints at appropriate scales). Morphisms are HCK–arrows preserving σ and projecting to the evident maps on J and B.

Proposition 9.2 (Bridge)**.** *The forgetful functors* $\mathsf{U}_{\mathrm{SK}} : \mathrm{HCK} \to \mathrm{SK}$ *and* $\mathsf{U}_{\mathrm{CK}} : \mathrm{HCK} \to \mathrm{CK}$ create *limits and* reflect *monomorphisms on the full subcategory of humans; hence the HCK category of humans is a concrete bridge between* SK *and* CK*.*

9.3 CK–driven hierarchy vs. SK–driven culture

CK delivers the *administrative* (system/political) hierarchy; SK sustains the *cultural/civilisational* mesh. Responsibilites separate accordingly:

R_1 (Work) governs tier hierarchy in CK; R_2, R_3, R_4 govern SK–cultural behaviour, thought, realisation.

A human administrator thus plays a *dual role*: when seated in a council he acts as an *administrator* (CK–political); otherwise he remains a *family member* (SK–cultural), continuously connected to all. This duality is enforced below by the "grounding in family" and "guided governance" constraints.

9.4 Level chain (tiers) and instance tree

Level chain (simple, canonical). We adopt the *chain-of-levels* model: define tier flags $\mathrm{W}_1, \ldots, \mathrm{W}_{10}$ as ledger–clopens (Def. 9.3) with parent map $p(k) = k + 1$; the root is W_{10}. This is a linear order on levels, not a branching of levels.

Definition 9.3 (Tier flags (canonical schema))**.** Using the R4/Scale schema with virtue regulators (unifying earlier variants):

$$
\begin{aligned}
&\mathrm{W}_1 := R_4 \wedge \mathrm{Sc}_1, && \mathrm{W}_2 := R_4 \wedge \mathrm{Sc}_2, && \mathrm{W}_3 := R_4 \wedge \mathrm{Sc}_2 \wedge V_j,\\
&\mathrm{W}_4 := R_4 \wedge \mathrm{Sc}_3, && \mathrm{W}_5 := R_4 \wedge \mathrm{Sc}_3 \wedge V_k, && \mathrm{W}_6 := R_4 \wedge \mathrm{Sc}_4,\\
&\mathrm{W}_7 := R_4 \wedge \mathrm{Sc}_4 \wedge V_\ell, && \mathrm{W}_8 := R_4 \wedge \mathrm{Sc}_5, && \mathrm{W}_9 := R_4 \wedge \mathrm{Sc}_5 \wedge V_{\ell'},\\
&\mathrm{W}_{10} := R_4 \wedge \mathrm{Sc}_5 \wedge V_{\ell''}.
\end{aligned}
$$

Each W_k is a finite union of ledger atoms (hence clopen) and is stable under all value nuclei by §8. A mapping-to-equivalence with the older Δ–sketch is recorded in Appendix F.

Instance tree (branching of *people/units*). At each level k the *instances* (families, villages, councils, . . .) form a branching forest; nomination/election induces upward edges to instances of level $k{+}1$. Formally, one can organise instances into a category Inst with a cloven opfibration $p : \mathrm{Inst} \to \mathbf{Lev}$ over the chain category $\mathbf{Lev} = \{1 \to \cdots \to 10\}$ (see Appendix H).

9.5 Roles and constraints in the Family tier

We fix roles and stage bounds inside W_1 (Family). These are ledger–clopen relations.

Definition 9.4 (Core roles in W_1)**.** *Child* $\hookrightarrow \mathrm{W}_1$ is the Stage–B subobject. *Parent Couple (PC)* is a ledger–clopen relation $\mathrm{PC} \hookrightarrow \mathrm{W}_1 \times \mathrm{W}_1$ with both members at least Stage–C. *Grandparents (GP)* are the PC of the PC, at least proto-D. *Spouse*, *Sibling* are the obvious fibre-product relations over the family. *Teacher (Guru)* is a Stage–D human *outside* W_1 with a teaching relation $\mathrm{Teach} \hookrightarrow \mathrm{W}_1 \times \mathrm{HCK}$ (pure SK–SK).

Constraint 9.5 (Stage–B minimality)**.** Stage–B humans occur only as *children* in W_1; never as PC/GP, nor in any W_k $(k \geq 2)$.

Constraint 9.6 (PC/GP stage bounds)**.** PC are at least Stage–C; GP are at least proto-D. These bounds are clopen and modal-stable.

Constraint 9.7 (Dual alignment for evolution)**.** Every Stage–B child $c \in \mathrm{W}_1$ is aligned (i) to a PC for physical/emotional support via $\mathrm{Align}^{\mathrm{phys}}(c, \mathrm{PC})$, and (ii) to a Stage–D Teacher for educational guidance via $\mathrm{Align}^{\mathrm{edu}}(c, \mathrm{Guru})$. Both alignments are necessary for $B \rightsquigarrow C$ (Paper 0).

Constraint 9.8 (Upward alignment)**.** PC align to their GP and to a Stage–D teacher; these alignments are in HCK (societal context), not inside the family object.

Remark 9.9 (Virtue flow and inherent asymmetry). Family (W_1) is inherently unequal: GP $\to$ PC $\to$ Child implements stewardship. The first three virtues (steadfastness, courage, generosity) flow along these CK–anchored links; the other three are transmitted via the SK relation Teach (teacher–student), independent of tiers.

9.6 Councils, stewardship, and guided governance

For each level k, let C_k be the council object and $\mathrm{Steward}_k \hookrightarrow C_k$ the *stewards* (ledger–clopen).

Definition 9.10 (Eligibility windows)**.** Stage–C humans are eligible for nomination/election up to W_3, proto–D up to W_5. Write $\mathrm{Elig}_C(k)$ and $\mathrm{Elig}_{\text{proto-D}}(k)$ for these windows.

Constraint 9.11 (Guidance by Stage–D)**.** Every steward has *constitutional guidance*: either is Stage–D or is aligned to a Stage–D Teacher via a ledger–clopen relation $\mathrm{Align}^{\mathrm{guide}} \hookrightarrow \mathrm{Steward}_k \times \mathrm{HCK}$, with the teacher outside W_1.

Definition 9.12 (Governance relation)**.** A *policy action* at level k is a subobject $\mathrm{PolRel}_k \hookrightarrow \mathrm{W}_1 \times C_k$ with fibres in R_1 (Work–responsibility), subject to two grounding axioms: (i) every administrator is anchored in some family (member of W_1); (ii) PolRel_k lands in *guided* stewards $\mathrm{Guided}_k \hookrightarrow \mathrm{Steward}_k$ (by Constraint 9.11).

Corollary 9.13 (Steward factorization principle)**.** *If every Stage–B child in* W_1 *enjoys both alignments* $\mathrm{Align}^{\mathrm{phys}}, \mathrm{Align}^{\mathrm{edu}}$*, then any policy* $\pi \subseteq \mathrm{PolRel}_k$ *that touches a child factors through a* guided *steward.* Proof. *Appendix G.*

Together, Constraints 9.7 and 9.11 ensure that all governance affecting children is necessarily mediated by guided stewards; see Appendix G.

9.7 Placement of slices relative to 10TWF

The slices cut across tiers. With $\int$ the decidable reflection:

$$\mathrm{SK} \;=\; \textstyle\int(\mathrm{W}_7), \qquad \mathrm{HCK} \;=\; \textstyle\int(\mathrm{W}_8), \qquad \mathrm{CK} \;=\; \textstyle\int(\mathrm{W}_{10}),$$

a convenient canonical placement consistent with applications; other consistent placements are possible.

Upgrade 9.1 (Higher-topos target)*.* All results in §9 use only finite limits, disjoint finite coproducts, images, and Lawvere–Tierney nuclei that preserve ledger–clopens. These features admit a 2-categorical lift in which the SSS refinement is implemented as a 2-nucleus on relations-between-relations (spans). Hence every 1-topos theorem in this section persists unchanged under the intended cohesive/$(\infty, 1)$ upgrade.

10 Related Work & Discarded Alternatives

This section situates our 27-generator, decidable-site construction within categorical logic and MD-style formalisms, and records approaches we prototyped but ultimately abandoned. We also document why the present path is *upgrade-safe* for the higher-topos programme without altering any 1-topos results proved here. Standard references for topos methods and site presentations include [4, 5].

10.1 Existing categorical models of value hierarchies

Author / school	**Core idea**	**Contrast with this paper**
Deligne (SGA $4\frac{1}{2}$) [6]	Booleanisation/double-negation of a coherent site.	Too strong for MD: on the *ledger–clopens* (our effective domain), Booleanisation makes value nuclei act trivially, erasing the modal content exploited in §8.
Joyal–Tierney / Johnstone–Moerdijk [2, 7, 8]	Hyperconnected–localic factorisations of geometric morphisms.	Hyperconnected stages inject many points, inflating generators and violating MD's "just-sufficient articulation." Our minimality (Thm. 4.4, Prop. 6.4) rules this out.
Vickers / Johnstone [9, 1]	Modalities as nuclei on locales (frames).	Conceptually close, but standard locale accounts rely on *infinitary* joins; our site is finitary (27 atoms; finite generating sieves).
Scholze (Condensed Mathematics) [14, 16]	Condensed sets (sheaves on compact Hausdorff spaces; profinite models are common) with rich analytic tools (bornologies, completed tensor products).	Ill-matched to discrete, decidable MD tiers; adds heavy structure without benefit to Morleyised decidability or the 27-atom ledger.
Bulk–boundary templates	Adjoint pairs linking a "bulk" to a "boundary."	Our comparison functor (Section 7) yields an *equivalence* of toposes; the geometric morphism $F \dashv F$ plays the role of a bulk–boundary bridge where the boundary datum is a *single* flag R_{walls}, not extra generators.

Remark. Historically, profinite/pro-étale cover technology motivates aspects of the condensed viewpoint; see [17]. Our site is deliberately *finite* and coherent to preserve decidability.

10.2 Higher-topos and $(\infty, 1)$ attempts

We prototyped an $(\infty, 1)$-topos where value layers are encoded by factorisation systems à la Lurie [13]; diagonals become n-truncations and Modal Transparency is recast as stability under Postnikov towers. *Failure*: coherence data propagates to an unbounded tower of k-cells, exploding the minimal ledger from 27 to $\geq \omega$ and breaking the finitary decidable brief (cf. also [2] for the general factorisation framework).

Upgrade 10.1 (Upgrade-safety summary). All core arguments in §§6–8 use only: (i) finite limits, (ii) *disjoint* finite coproducts, (iii) images, and (iv) Lawvere–Tierney topologies presented as idempotent, left-exact monads that preserve *ledger–clopens*. These features admit a 2-categorical lift where SSS becomes a 2-nucleus (relations between relations). Hence the 1-topos results here are stable under the intended higher-topos upgrade (cf. Upgrade 9.1).

10.3 Discarded Morleyisation variants

Variant	# flags	Why rejected
Five-flag (early)	Walls, Virtues, Responsibilities, Scales, Shells	Redundant—A4/A6/A8 already make three diagonals clopen and Shells are nested (A9). Raises the ledger to 31 for no gain.
Two-flag (mid-draft)	Walls + Scales	Scales are already decidable by (A7)–(A8); violates minimality (Thm. 4.4).
Zero-flag + weakened exclusivity	0	Leaves the Wall diagonal undecidable; equality fails to split the loop in proof-search.

Verdict. The single-flag solution (Walls only) is the *only* variant meeting all three constraints: decidability, philosophical fidelity, and minimal ledger size (Sections 4–5).

10.4 Why not Booleanise at the end?

A common suggestion—"just Booleanise"—fails for our aims:

- *Modal content collapses on the ledger.* In a Boolean setting, the five value nuclei act trivially on *ledger–clopens*, eliminating the analytic–synthetic signal we need (cf. §8).
- *Tier hierarchy flattens.* Complements become uniform; the ten tier flags lose informative separation.
- *Philosophical mismatch.* MD models *growth* of certainty; Booleanisation imposes global certainty by fiat, not by value-layer closure.

10.5 Summary of the chosen path

- **Minimal extension**. One Morley flag R_{walls}; nothing more.
- **Finite, decidable site**. 27 generators; **four** generating cover sieves (Walls, Virtues, Responsibilities, Scales); Shells nested only. (Prop. 6.4, App. C.)
- **Heyting logic intact**. Value nuclei act non-trivially and preserve complements on ledger–clopens (Thm. 8.4).
- **Topos-level faithfulness**. Equivalence to the syntactic topos $\mathscr{W}(T_{\text{MD}})$ (Section 7).
- **Modal & tier alignment**. SK–HCK–CK slices and 10TWF tiers are realised without extra gadgets and are upgrade-safe (Section 9, Upgrade 10.1).

All alternative routes either added superfluous structure, undermined MD's interpretation, or destroyed decidability. The construction presented here is therefore the simplest *and* most faithful categorical realisation of Madhyasth Darshan to date, engineered to lift to higher toposes without revising any theorem proved in this paper.

11 Conclusion and Outlook

11.1 Achievements

- **Minimal, decidable foundation.** A single Morley flag R_{walls} converts the Madhyasth–Darshan axiom set into a coherent, *decidable* site with 27 generators and *four* generating cover sieves (Walls, Virtues, Responsibilities, Scales), Shells nested only. All unary diagonals on *ledger–clopens* are clopen (Sections 4–6).

- **Conservativity and transport.** The Morley step is a *definitional* (conservative) extension on formulas mentioning Walls; proofs transport functorially between the original and Morleyised theories (Section 5).

- **Topos-level faithfulness.** Section 7 proves $\mathrm{Sh}\big(\mathcal{C}^{*}_{\mathrm{MD}}, \mathcal{J}^{*}\big) \simeq \mathscr{W}\big(T_{\mathrm{MD}}\big)$, so no logical content is lost in passing to the small, decidable universe.

- **Modal transparency.** The five value nuclei $\square_J, \square_D, \square_b, \square_r, \square_\ell$ (idempotent, left-exact *monads* on the ambient topos) preserve complements on ledger–clopens (Theorem 8.4); hence decidability ascends every modal layer (Section 8).

- **World–Family realised and sliced.** The ten tier flags $W_1, \ldots, W_{10}$ are clopen and stable under all nuclei, and the knowledge slices embed as conservative opens in the quantitative order CK $\subset$ HCK $\subset$ SK (Section 9), with qualitative adequacy ordered SK $\succcurlyeq$ HCK $\succcurlyeq$ CK.

- **Logical economy validated.** Alternatives—Booleanisation, multi-flag Morleyisation, or direct $(\infty, 1)$ encodings—either collapse modal content, break minimality, or destroy decidability; the single-flag path is both simplest and most faithful (Section 10).

- **Upgrade-safety.** All core arguments use only finite limits, *disjoint* finite coproducts, images, and nuclei stable on ledger–clopens, so the 1-topos results lift to the planned higher-topos treatment of "relations-between-relations" SSS without revision (Upgrades 9.1, 10.1).

11.2 Open problems and research directions

Theme	Question	Motivation
Higher SSS (relation-of-relations)	Promote the family-scoped SSS to a 2-nucleus; characterise modal transport at 2-level.	Completes co-existence as higher-topos structure while preserving 1-topos theorems.
Iterated value cubes	Can the five nuclei assemble into an iterated tense-modal cube *on ledger–clopens* without losing decidability?	Models compound MD virtues (e.g. truth-in-responsibility) with guaranteed crispness.
Formal verification	Mechanise §§7–9 in a proof assistant; publish an add-on.	Delivers machine-checked credibility and a reusable categorical library.
Socio-ecological modelling	Use the 10-Tier flags as state variables in agent-based simulations.	Empirically test MD predictions at planetary scale.

11.3 Closing remark

A single logical refinement renders Madhyasth Darshan's layered ontology *computably crisp* while preserving its intuitionistic spirit. The resulting 27-atom, ledger-driven universe is mathematically rigorous, ready for mechanised proof, and engineered to lift—unchanged in its core theorems—to a higher-topos expression of co-existence as "relations-between-relations."

A Boolean ledger on the geometric atoms

Let $\mathcal{A} = \{$the 26 geometric atoms $\Sigma_1, \ldots, \Sigma_6,\ V_1, \ldots, V_6,\ R_1, \ldots, R_4,\ \mathrm{Sc}_1, \ldots, \mathrm{Sc}_5,\ \Delta_1, \ldots, \Delta_5\}$. By the family-scoped SSS development, these atoms are idempotent, pairwise disjoint *within each family*, Shells form a nested chain, and the union of all 26 atoms covers truth; see [20]. In this appendix we work with the Boolean subalgebra

$$\Omega_{\mathcal{A}} \ := \ \Big\{ \bigvee_{a \in S} a \ \Big|\ S \subseteq \mathcal{A} \Big\} \ \subseteq \ \Omega,$$

the *ledger-definable clopens* generated by the 26 geometric atoms.[3]

Two canonical maps

Define

$$F : \Omega_{\mathcal{A}} \longrightarrow \mathbf{2}^{\mathcal{A}}, \qquad F(x) := \{\, a \in \mathcal{A} \mid a \leq x \,\},$$
$$G : \mathbf{2}^{\mathcal{A}} \longrightarrow \Omega_{\mathcal{A}}, \qquad G(S) := \bigvee_{a \in S} a.$$

Lemma A.1 (Homomorphism laws)**.** *F preserves finite meets and bounds; G preserves finite joins and bounds.*

Proof. For F: $a \leq x \wedge y$ iff $a \leq x$ and $a \leq y$, hence $F(x \wedge y) = F(x) \cap F(y)$, and similarly for $\top, \bot$. For G: $\bigvee_{a \in S \cup T} a = (\bigvee_{a \in S} a) \vee (\bigvee_{a \in T} a)$; bounds are immediate. □

Lemma A.2 (Section–retraction)**.** *$F \circ G = \mathrm{id}_{\mathbf{2}^{\mathcal{A}}}$.*

Proof. For $S \subseteq \mathcal{A}$, the atoms in $\mathcal{A}$ are idempotent and pairwise disjoint within each family, so $a \leq \bigvee_{b \in S} b$ iff $a \in S$. Thus $F(G(S)) = S$. □

Theorem A.3 (Boolean completeness of the geometric ledger)**.** *G and F are mutually inverse lattice isomorphisms between $\mathbf{2}^{\mathcal{A}}$ and $\Omega_{\mathcal{A}}$. In particular, every ledger-definable element is* clopen*, with complement*

$$x^* = G\big(\mathcal{A} \setminus F(x)\big).$$

Proof. Lemma A.2 gives $F \circ G = \mathrm{id}$. For $x \in \Omega_{\mathcal{A}}$ write $x = \bigvee_{a \in S} a$; then $G(F(x)) = G(S) = x$. Combine with Lemma A.1. □

Remark A.4 (Analytic–synthetic duality)*.* The isomorphism $\Omega_{\mathcal{A}} \cong \mathbf{2}^{\mathcal{A}}$ realises the MD epistemic cycle: F *analyses* a truth-value into its atomic profile, while G *re-synthesises* that profile. Because $G \circ F = \mathrm{id}$ and $F \circ G = \mathrm{id}$, analysis and synthesis are mutually verifying: every ledger-definable proposition is reconstructible from its 26-atom footprint.

[3] The Morley flag R_{walls} is a logical name for $\Sigma_1 \vee \cdots \vee \Sigma_6$ (A11a–b) and is *not* counted among the geometric atoms; it is unnecessary for this appendix.

Corollaries

Corollary A.5 (Faithful embedding)**.** *$F : \Omega_{\mathcal{A}} \to \mathbf{2}^{\mathcal{A}}$ is fully faithful; the atom-profile is a complete invariant.*

Corollary A.6 (Self-duality)**.** *G is left adjoint to F; the adjunction is an equivalence on $\Omega_{\mathcal{A}}$.*

Corollary A.7 (Constructive complement)**.** *For $x \in \Omega_{\mathcal{A}}$, the complement is $x^* = G(\mathcal{A} \setminus F(x))$ (no choice used).*

Remark. A 27-atom variant (including R_{walls}) is proved in Paper A, Appendix A [23].

B Proofs of Irreducibility, Sufficiency, and Diagonal–split

Cited in §3; applies to Sections 4–5 and used in §§6–7.

Foundational note. All arguments here are intuitionistic and internal to the coherent syntactic category; we use only finite limits, disjoint finite coproducts, and stability of pullbacks of coproduct injections. Background: [4, Chs. III–IV], [5, Ch. 2], [1, §C2].

A. Necessity (Lemma 4.2)

Lemma (Singleton-flag irreducibility)*.* If no new predicate is added to the language Σ_{MD}, the Wall family-union diagonal is not complemented; hence at least one extra predicate is necessary.

Proof. Work in the coherent syntactic category of T_{MD} (A0–A9, with A2 *omitted*). Write $R_\Sigma(x) := \Sigma_1(x) \vee \cdots \vee \Sigma_6(x)$. Assume for contradiction that there is a geometric complement $U \hookrightarrow R_\Sigma \times R_\Sigma$ of Δ_{R_Σ} provable from T_{MD}.

Consider the explicit Set-model $\mathcal{M}$ with carrier $\{*\}$ and interpretation $\Sigma_p(*) = \Sigma_q(*) = \top$ for fixed $p \neq q$, all other $\Sigma_i(*) = \bot$; interpret the other families so that A3–A9 hold (possible on a singleton). Then $\mathcal{M} \models R_\Sigma(*)$, and $(*, *) \in \Delta_{R_\Sigma}$. But any basic geometric rectangle containing $\Sigma_p(u) \wedge \Sigma_q(v)$ also contains $(*, *)$, so every geometric U that covers all off-diagonal pairs meets Δ_{R_Σ} in $\mathcal{M}$, contradicting complementarity. Hence no geometric complement of Δ_{R_Σ} is provable in T_{MD}. □

MD alignment. The unresolved Wall ambiguity cannot be eliminated without *naming* it; the Morley flag R_{walls} is the minimal articulation.

B. Sufficiency (Lemma 4.3)

Lemma (Sufficiency)*.* Adding the single flag R_{walls} (A11a–b) and exclusivity (A2*) complements every previously undecidable diagonal.

Proof. With A2* the family $\{\Sigma_i\}_{i=1}^6$ is pairwise disjoint; A11a–b yields an isomorphism

$$R_{\text{walls}} \;\cong\; \Sigma_1 \sqcup \cdots \sqcup \Sigma_6$$

in the syntactic category. Stability of finite coproducts under pullback gives a canonical decomposition

$$R_{\text{walls}} \times R_{\text{walls}} \;\cong\; \bigsqcup_{i,j} (\Sigma_i \times \Sigma_j).$$

Under this isomorphism, $\Delta_{R_{\text{walls}}}$ identifies with $\bigsqcup_k (\Sigma_k \times \Sigma_k)$, and the clopen complement

$$U \;:=\; \bigsqcup_{i \neq j} (\Sigma_i \times \Sigma_j)$$

is disjoint from $\Delta_{R_{\text{walls}}}$ and satisfies $\Delta_{R_{\text{walls}}} \vee U = R_{\text{walls}} \times R_{\text{walls}}$. Thus $\Delta_{R_{\text{walls}}}$ is complemented. The same argument applies to any finite disjoint union. □

MD alignment. Once the Wall state is *decidably named*, all equalities on recognisable states (ledger–clopens) become decidable.

C. Diagonal–split lemma (full proof of Lemma 5.1)

Lemma (Diagonal–split template). If $R \cong \Sigma_1 \sqcup \cdots \sqcup \Sigma_n$ is a finite disjoint union in a coherent theory, then $\Delta_R \subseteq R \times R$ is complemented.

Proof. Let $i_k : \Sigma_k \to R$ be the coproduct injections. Disjointness gives pullbacks $\Sigma_k \cong R \times_R \Sigma_k$ and $0 \cong \Sigma_i \times_R \Sigma_j$ for $i \neq j$. By stability of coproducts under pullback,

$$R \times R \;\cong\; \Big(\bigsqcup_k \Sigma_k\Big) \times \Big(\bigsqcup_\ell \Sigma_\ell\Big) \;\cong\; \bigsqcup_{k,\ell} (\Sigma_k \times \Sigma_\ell).$$

The summands with $k = \ell$ form (the image of) the diagonal Δ_R, while those with $k \neq \ell$ define a clopen subobject U disjoint from Δ_R. Hence Δ_R is complemented by U. □

MD alignment. Finite, mutually exclusive recognisable states (once named) support crisp equality—exactly the behaviour required on the ledger–clopens.

Dependencies. A11a–b (Morley flag) and A2* (exclusivity) are used only in B; A0–A9 (with A2 omitted) are used in A; C relies solely on finitary coproduct calculus (no classical axioms).

C Technical facts for the site $(\mathcal{C}^*_{\text{MD}}, J^*)$

C.1 Coherent envelope preserves generators

Proposition C.1. *Let $\mathcal{C}^*_0$ be the preorder of literals and finite meets from §subsec:site-objects. Its coherent envelope $\mathcal{C}^*_{\text{MD}}$ (free completion by finite limits, stable images, and finite coproducts) introduces no new indecomposable generators: every literal remains indecomposable, i.e. if a literal ℓ is isomorphic to $X \vee Y$ in $\mathcal{C}^*_{\text{MD}}$, then $X \simeq \bot$ or $Y \simeq \bot$.*

Proof. Finite coproducts in a coherent category are disjoint and universal; the inclusion $\ell \hookrightarrow X \vee Y$ factors through one summand by indecomposability of literals in $\mathcal{C}^*_0$, forcing the other to be initial. Universality transports this to $\mathcal{C}^*_{\text{MD}}$. □

C.2 Joint epimorphy of generating sieves

Lemma C.2. *The families* $\{V_j \to \top\}_{j=1}^{6}$, $\{R_k \to \top\}_{k=1}^{4}$, $\{\mathrm{Sc}_\ell \to \top\}_{\ell=1}^{5}$, *and* $\{\Sigma_i \to R_{walls}\}_{i=1}^{6}$ *are jointly epimorphic in* $\mathcal{C}^*_{\mathrm{MD}}$.

Proof. For V, R, Sc: (A3)/(A5)/(A7) give the covers, and (A4)/(A6)/(A8) give pairwise disjointness, so any arrow into $\top$ factors locally through one inclusion. For Walls: (A11a–b) identify R_{walls} with the union of Σ_i, while A2* gives disjointness; the same argument applies. □

C.3 Subcanonicity

Lemma C.3. *Every representable presheaf on* $(\mathcal{C}^*_{\mathrm{MD}}, J^*)$ *is a* J^**-sheaf; hence* J^* *is subcanonical.*

Proof. Each generating sieve is induced by a provable cover-sequent in T^*_{MD}. Provability is stable under substitution/pullback, so the sheaf condition holds for representables. □

C.4 Properties used in the density proof

Lemma C.4 (Pullback stability of covers)**.** *If a sieve* S *on* X *is* J^**-covering and* $f : Y \to X$*, then the pullback* f^*S *is* J^**-covering.*

Proof. Covering is defined by a provable sequent in T^*_{MD}; substitution along f preserves provability in the coherent envelope. □

Lemma C.5 (Finite sums of covering sieves)**.** *If* S *covers* A *and* T *covers* B*, then the induced sieve covers* $A \vee B$*. More generally, a finite family of covering sieves* $\{S_i \text{ on } A_i\}$ *yields a covering sieve on* $\bigvee_i A_i$*.*

Proof. Finite coproducts are universal and disjoint; cover checks are local and preserved by the coproduct's universal property. □

Lemma C.6 (Closure under finite limits)**.** *If* S *covers* A *and* T *covers* B*, then after pulling back along the projections, the product sieve covers* $A \times B$*.*

Proof. By Lemma C.4, covers are stable under pullback; apply to projections and take the product family. □

Lemma C.7 (Images preserve joint epimorphy)**.** *If* $\{u_i : U_i \to X\}$ *is* J^**-covering, then the family of regular epis* $\{\mathrm{im}(u_i) \to X\}$ *is* J^**-covering.*

Proof. Each u_i factors as $U_i \twoheadrightarrow \mathrm{im}(u_i) \hookrightarrow X$. Thus any arrow factoring locally through u_i also factors through its image. Moreover the image of $\bigvee_i U_i \to X$ is $\bigvee_i \mathrm{im}(u_i) = X$ (images distribute over finite coproducts in a coherent category), so the images remain jointly epimorphic. □

Proposition C.8 (Closure for the density argument)**.** *Let* $\mathcal{C} \subseteq \mathcal{C}^*_{\mathrm{MD}}$ *be the full subcategory of objects* Y *for which there exists a* J^**-cover of* Y *by arrows in the image of the functor* $F : \mathrm{Syn}(T_{\mathrm{MD}}) \to \mathcal{C}^*_{\mathrm{MD}}$*. Then* $\mathcal{C}$ *contains the* 27 *generators and is closed under finite limits, finite coproducts, and images. Hence* $\mathcal{C} = \mathcal{C}^*_{\mathrm{MD}}$*.*

Proof. Generators are covered by Lemma 7.1 in the main text. Closure under finite limits follows from Lemma C.4 applied to projections. Closure under finite coproducts follows from Lemma C.5. Closure under images follows from Lemma C.7. Since $\mathcal{C}^*_{\mathrm{MD}}$ is generated from its 27 atoms by these operations (Proposition C.1), every object lies in $\mathcal{C}$. □

D Philosophical Grounding of the 27-Atom Ledger

The 26+1 predicates constitute a minimal, sufficient ledger for core classifications in Madhyasth Darshan (MD) [18, 19]. Family-scoped orthogonality, nesting, and non-redundancy are *realised* by the SSS construction in [20] (with representation-theoretic provenance in [21]). In the main text we present A1–A9 as geometric axioms to keep the site finitary and decidable; Walls exclusivity re-enters only after Morleyisation (A2*). *Notation:* we write $\neg R_{\text{walls}} := \neg R_{\text{walls}}$.

Shells Δ: constitution of the self (nesting)

- Δ_1 Atma, Δ_2 Buddhi, Δ_3 Chitta, Δ_4 Vritti, Δ_5 Mana.
- *Formal link:* A9 encodes concentric nesting $\Delta_{m+1} \vdash \Delta_m$ $(m = 1, \ldots, 4)$.

Responsibilities R: arenas of humane conduct (exclusive)

- R_1 Realisation, R_2 Thought, R_3 Behaviour, R_4 Work.
- *Formal link:* A5 (cover), A6 (exclusivity = non-substitutability).

Scales Sc: contexts of living (exclusive)

- Sc_1 Self, Sc_2 Family, Sc_3 Society, Sc_4 Nature, Sc_5 Co-existence.
- *Formal link:* A7 (cover), A8 (exclusivity).

Virtues V: intrinsic nature at the Vritti layer (exclusive)

- V_1 Fortitude (*dheerta*), V_2 Courage (*veerta*), V_3 Generosity (*udaarta*), V_4 Kindness (*daya*), V_5 Grace (*krupa*), V_6 Compassion (*karuna*).
- *Formal link:* A3 (cover), A4 (exclusivity).

Walls Σ: agitations of outward-oriented Mana

- Σ_1 Lust, Σ_2 Anger, Σ_3 Intoxication, Σ_4 Infatuation, Σ_5 Greed, Σ_6 Wrath.
- *Formal link:* A1 (cover). Walls exclusivity is *deferred* in T_{MD} and reinstated after Morleyisation as A2*; A11a–b introduces the Morley flag R_{walls} with $R_{\text{walls}} \dashv\vdash \Sigma_1 \vee \cdots \vee \Sigma_6$.

"Folding back" of Mana and the role of R_{walls}

Awakening reverses the outward flow of *Mana* and orients it to inner layers (Vritti $\to$ Chitta $\to$ Buddhi $\to$ Atma). In the ledger:

- The Morley flag R_{walls} *names the union* $\Sigma_1 \vee \cdots \vee \Sigma_6$ (A11a–b); it is true exactly when a Wall-state is present.
- The "resolved" condition is the *complement* $\neg R_{\text{walls}} := \neg R_{\text{walls}}$, which exists and is clopen in the decidable site (Lemma 5.1, Cor. 5.2).
- Stage–D entails $\neg R_{\text{walls}}$ and modal stability; we do not collapse Stage–D to a single predicate to keep the ledger minimal.

Quick reference (optional).

Virtues V	
V_1	Fortitude (*dheerta*)
V_2	Courage (*veerta*)
V_3	Generosity (*udaarta*)
V_4	Kindness (*daya*)
V_5	Grace (*krupa*)
V_6	Compassion (*karuna*)

Walls Σ	
Σ_1	Lust
Σ_2	Anger
Σ_3	Intoxication
Σ_4	Infatuation
Σ_5	Greed
Σ_6	Wrath

Where this is used. Axioms (Sections 2–3); Morley flag and decidability (Sections 4–5); modal stability (Section 8); 10TWF tiers (Section 9).

E Catalogue of Tier Flags and Role Relations

Logic and scope. All formulæ below are *ledger–clopen*: they are built from finite Boolean combinations of the 26 geometric atoms (Walls, Virtues, Responsibilities, Scales, Shells) and hence are stable under all five value nuclei (§8). MD alignment: tiers/selectors isolate recognisable states (vivek/samanvay cycle) without collapsing modalities.

E.1 Tier flags: concrete MSC realisation

We adopt the canonical R4/Scale schema from Def. 9.3 and instantiate the virtue regulators by a fixed MSC table:

Tier k	$V^{(k)} \subseteq \{1, \dots, 6\}$
3	$\{1\}$ (Fortitude)
5	$\{2\}$ (Courage)
7	$\{3\}$ (Generosity)
9	$\{4, 5\}$ (Kindness or Grace)
10	$\{6\}$ (Compassion)

(First three virtues flow along CK-anchored family links; the remaining three are transmitted via the SK teaching relation, cf. Remark 9.9.)

$$
\begin{aligned}
\mathrm{W}_1 &:= R_4 \wedge \mathrm{Sc}_1, \\
\mathrm{W}_2 &:= R_4 \wedge \mathrm{Sc}_2, \\
\mathrm{W}_3 &:= R_4 \wedge \mathrm{Sc}_2 \wedge \bigvee_{i \in V^{(3)}} V_i \;=\; R_4 \wedge \mathrm{Sc}_2 \wedge V_1, \\
\mathrm{W}_4 &:= R_4 \wedge \mathrm{Sc}_3, \\
\mathrm{W}_5 &:= R_4 \wedge \mathrm{Sc}_3 \wedge \bigvee_{i \in V^{(5)}} V_i \;=\; R_4 \wedge \mathrm{Sc}_3 \wedge V_2, \\
\mathrm{W}_6 &:= R_4 \wedge \mathrm{Sc}_4, \\
\mathrm{W}_7 &:= R_4 \wedge \mathrm{Sc}_4 \wedge \bigvee_{i \in V^{(7)}} V_i \;=\; R_4 \wedge \mathrm{Sc}_4 \wedge V_3, \\
\mathrm{W}_8 &:= R_4 \wedge \mathrm{Sc}_5, \\
\mathrm{W}_9 &:= R_4 \wedge \mathrm{Sc}_5 \wedge \bigvee_{i \in V^{(9)}} V_i \;=\; R_4 \wedge \mathrm{Sc}_5 \wedge (V_4 \vee V_5), \\
\mathrm{W}_{10} &:= R_4 \wedge \mathrm{Sc}_5 \wedge \bigvee_{i \in V^{(10)}} V_i \;=\; R_4 \wedge \mathrm{Sc}_5 \wedge V_6.
\end{aligned}
$$

Family audit per flag. Each W_k uses Responsibilities (R_4) and a single Scale; the virtue family appears exactly for $k \in \{3, 5, 7, 9, 10\}$ as indicated. Walls and Shells do not occur in the definitions (their effects are mediated via $\neg R_{\text{walls}}$ and capacity lemmas).

E.2 Roles inside W_1

The following are ledger–clopen *relations* in the Family tier; stage bounds are added as clopen constraints in SK (cf. §9.5).

Definition E.1 (Core role relations in Family)**.**

- *Couple (PC)* is a symmetric subobject $\mathsf{PC} \hookrightarrow \mathrm{W}_1 \times \mathrm{W}_1$ (ledger–clopen).
- *Child–of–PC* is a subobject $\mathsf{ChildOf} \hookrightarrow \mathrm{W}_1 \times (\mathrm{W}_1 \times \mathrm{W}_1)$ with codomain restricted to PC. Write $\mathsf{ParentOf}(p, c)$ for $\exists q.\ \mathsf{ChildOf}(c, (p, q))$.
- *Child* is the unary subobject $\mathsf{Child} \hookrightarrow \mathrm{W}_1$ defined by $\mathsf{Child}(c) := \exists p, q.\ \mathsf{ChildOf}(c, (p, q))$. (Stage–B is an additional clopen bound, cf. Constraint 9.5.)
- *Grandparent (GP)* is the binary relation $\mathsf{GP}(g, c) := \exists p.\ \mathsf{ParentOf}(g, p) \wedge \mathsf{ParentOf}(p, c)$. (Add the clopen stage bound proto-D(g) as in Constraint 9.6.)
- *Spouse* is $\mathsf{Spouse}(x, y) := \mathsf{PC}(x, y)$.
- *Sibling* is $\mathsf{Sibling}(x, y) := \exists p, q.\ \mathsf{ChildOf}(x, (p, q)) \wedge \mathsf{ChildOf}(y, (p, q)) \wedge (x \neq y)$.

Stage bounds (Child = Stage–B, PC members $\geq$ Stage–C, GP $\geq$ proto-D) are recorded in Constraints 9.5–9.6; see §9.5. MD alignment: stewardship GP→PC→Child implements value flow; early virtues (1–3) follow these CK-anchored links, later virtues (4–6) propagate via teaching.

E.3 Cross-tier relations

Definition E.2 (Friendship, Work, Teach)**.**

- *Friendship* is a symmetric SK–relation $\mathsf{Friend} \hookrightarrow \mathrm{W}_1 \times \mathrm{W}_1$ with fibres contained in $\neg R_{\text{walls}}$ on both arguments; optionally gate by $V_4 \vee V_5 \vee V_6$.

- *Work* is the HCK–relation $\mathsf{Work} \hookrightarrow \left(\bigvee_k \mathrm{W}_k\right) \times \left(\bigvee_k \mathrm{W}_k\right)$ whose fibres lie in R_1 (Work–responsibility).
- *Teach* is the SK–relation $\mathsf{Teach} \hookrightarrow \mathrm{W}_1 \times \mathrm{HCK}$ with the second component a Stage–D Teacher (outside W_1); cf. Def. 9.4.

E.4 Councils and stewards

Definition E.3 (Councils and guided stewards)**.** For each level k, let C_k be the *council* object. The subobject $\mathsf{Steward}_k \hookrightarrow C_k$ picks *stewards* (ledger–clopen). Guided stewards form $\mathsf{Guided}_k \hookrightarrow \mathsf{Steward}_k$ defined by the clopen "guidance" alignment (either Stage–D or aligned to a Stage–D Teacher), cf. Constraint 9.11.

Family audit. $\mathsf{PC}, \mathsf{ChildOf}, \mathsf{ParentOf}, \mathsf{Sibling}, \mathsf{Spouse}$ use only W_1 and equality; Friend (NoWall, $V_{4\vee 5\vee 6}$) uses Virtues/Walls; Work uses R_1; Teach is pure SK with Tier typing; $C_k, \mathsf{Steward}_k, \mathsf{Guided}_k$ are relations/objects over tiers.

MD alignment note. The above relations formalise humane co-existence: family stewardship (CK) linked to teaching/culture (SK), with all selectors ledger–clopen so value nuclei preserve decisional crispness.

F Equivalence of Δ–Scheme and R4/Scale Tier Scheme

We compare the older Δ–based sketch $T_k^{(\Delta)}$ with the canonical $T_k^{(R_4,\mathrm{Sc},V)}$ of Def. 9.3.

Proposition F.1 (Interdefinability up to j–closure)**.** *For each k, $T_k^{(\Delta)}$ and $T_k^{(R_4,\mathrm{Sc},V)}$ are interdefinable as ledger–clopens (up to j–closure). Shell nesting (A9) and SSS capacity bounds suffice to transport membership both ways.*

Proof. Step 1 (Normal form for shells). By A9 the shell literals form a chain $\Delta_{m+1} \vdash \Delta_m$. Any Δ–conjunctive condition appearing in $T_k^{(\Delta)}$ reduces to a single window $\Delta_{\geq m} := \Delta_m$ by contraction; finite unions of such windows remain clopen.

Step 2 (Capacity map $\Delta \Rightarrow \mathrm{Sc}$*).* The SSS capacity bounds assign to each shell window $\Delta_{\geq m}$ a minimal viable scale $\mathrm{Sc}_{\mathrm{cap}(m)}$ (monotone in m). Replace the Δ guard by $\mathrm{Sc}_{\mathrm{cap}(m)}$ to obtain a formula of the form $R_4 \wedge \mathrm{Sc}_\bullet$ (and, where present in $T_k^{(\Delta)}$, add the virtue regulator V_i from the MSC table for that tier). This yields $T_k^{(R_4,\mathrm{Sc},V)}$. All replacements use finite joins/meets, hence preserve clopenness.

Step 3 (Back translation $\mathrm{Sc} \Rightarrow \Delta$*).* Conversely, given $T_k^{(R_4,\mathrm{Sc},V)}$, pick any shell window $\Delta_{\geq m}$ with $\mathrm{cap}(m) \leq \bullet$. Then $R_4 \wedge \mathrm{Sc}_\bullet$ implies $R_4 \wedge \Delta_{\geq m}$ and the virtue guard is copied verbatim. Finite unions over admissible m produce an equivalent Δ–presentation.

Step 4 (Closure). Where either translation introduces a harmless refinement, equality holds up to j–closure in the ambient topos (value nuclei are left exact and preserve clopens), so both sides classify the same subobject. □

MD alignment note. The Δ–scheme expresses constitution (inner layering), while the $R_4/\mathrm{Sc}/V$ scheme expresses conduct/context regulators; their equivalence reflects MD's "analysis–synthesis" unity at the tier level.

G Worked Proof of the Steward Factorization Principle

(We give the detailed proof of Cor. 9.13 here.)

Foundational stance. All arguments are intuitionistic and internal to the ambient topos. We use: finite limits, disjoint finite coproducts, images (regular epi/mono factorisation), and that the five value nuclei are idempotent left-exact monads preserving ledger–clopens (cf. §8).

Setup and notation

Fix a level k. Let C_k be the council object, with the *steward* subobject $\mathsf{Steward}_k \hookrightarrow C_k$ and the *guided steward* subobject $\mathsf{Guided}_k \hookrightarrow \mathsf{Steward}_k \hookrightarrow C_k$. By definition of Guided_k (see §9), there is a clopen predicate $\mathrm{Align}_k(c)$ on C_k such that

$$\mathsf{Guided}_k \;=\; \mathsf{Steward}_k \wedge \mathrm{Align}_k \qquad \text{and} \qquad \mathrm{Align}_k(c) \;\equiv\; \big(\mathrm{Stage\text{–}D}(c) \vee \exists t.\ \mathrm{Teacher}(t) \wedge \mathsf{Teach}(t,c)\big),$$

with Stage–D, Teacher, and Teach the (ledger–clopen) stage and teaching predicates/relations from §9. In particular Guided_k is ledger–clopen and hence preserved by all value nuclei.

Given a morphism $f : X \to C_k$, write $\Gamma_f \hookrightarrow X \times C_k$ for its graph, i.e. the subobject classified by $(x, c) \mapsto (f(x){=}c)$.

Main factorisation

Lemma G.1 (Image containment)**.** *Let $f : X \to C_k$ be a morphism whose graph satisfies $\Gamma_f \le X \times \mathsf{Steward}_k$ and $\Gamma_f \le X \times \mathrm{Align}_k$. Then $\mathrm{im}(f) \le \mathsf{Guided}_k$ as subobjects of C_k.*

Proof. Work internally. Take any $c \in \mathrm{im}(f)$. By definition of image, $\exists x.\ f(x) = c$. Because $\Gamma_f \subseteq X \times \mathsf{Steward}_k$, we have $\mathsf{Steward}_k(c)$; and because $\Gamma_f \subseteq X \times \mathrm{Align}_k$, we have $\mathrm{Align}_k(c)$. Hence $\mathsf{Guided}_k(c)$, so $c \in \mathsf{Guided}_k$. Thus $\mathrm{im}(f) \le \mathsf{Guided}_k$. □

Proposition G.2 (Steward factorisation)**.** *Under the hypotheses of Lemma G.1, there exists a unique $\overline{f} : X \to \mathsf{Guided}_k$ with*

$$\iota \circ \overline{f} \;=\; f,$$

where $\iota : \mathsf{Guided}_k \hookrightarrow C_k$ is the inclusion. Moreover, $\overline{f}$ is nucleus-stable in the sense that its graph is ledger–clopen (hence fixed by each value nucleus).

Proof. Factor f as a regular epi followed by a mono: $f = m \circ e$ with $e : X \twoheadrightarrow \mathrm{im}(f)$ and $m : \mathrm{im}(f) \hookrightarrow C_k$. By Lemma G.1, m refines through the inclusion $\iota : \mathsf{Guided}_k \hookrightarrow C_k$, i.e. there is a unique mono $j : \mathrm{im}(f) \hookrightarrow \mathsf{Guided}_k$ with $\iota \circ j = m$. Put $\overline{f} := j \circ e : X \to \mathsf{Guided}_k$. Then $\iota \circ \overline{f} = \iota \circ j \circ e = m \circ e = f$.

Uniqueness: if $\iota \circ g = f$ then, by monocity of ι, the graph of g equals the pullback of Γ_f along $\mathrm{id}_X \times \iota$, hence g equals $\overline{f}$.

For nucleus stability: Guided_k is ledger–clopen and value nuclei are left-exact; thus the classifying subobject of $\Gamma_{\overline{f}}$ is preserved by each nucleus. □

From relations to functions

Often one starts from a ledger–clopen *relation* $R \hookrightarrow X \times C_k$ expressing "x is guided by c" rather than a map f. The following standard functionalisation reduces this to Proposition G.2.

Lemma G.3 (Functionalisation). *Suppose $R \hookrightarrow X \times C_k$ is ledger–clopen, total and single-valued:*

$$\forall x\, \exists! c.\ R(x, c).$$

Then there is a unique $f : X \to C_k$ with $\Gamma_f = R$. If moreover $R \ \leq \ X \times \mathsf{Steward}_k$ and $R \ \leq \ X \times \mathrm{Align}_k$, then f satisfies the hypotheses of Lemma G.1.

Proof. The existence/uniqueness of f with $\Gamma_f = R$ is standard in any elementary topos: totality gives existence; single-valuedness gives uniqueness. The extra assumptions imply $\Gamma_f \subseteq X \times \mathsf{Steward}_k$ and $\Gamma_f \subseteq X \times \mathrm{Align}_k$ by definition, so Lemma G.1 applies. □

Conclusion: proof of Cor. 9.13

Proof of Cor. 9.13. Instantiate X with the stated domain (e.g. a union of tiers), take R to be the given guidance relation (ledger–clopen, nucleus-stable), and check the two side-conditions:

$$R \ \leq \ X \times \mathsf{Steward}_k \qquad \text{and} \qquad R \ \leq \ X \times \mathrm{Align}_k.$$

By Lemma G.3 we obtain $f : X \to C_k$ with graph R, and by Proposition G.2 a unique factorisation $X \xrightarrow{\overline{f}} \mathsf{Guided}_k \hookrightarrow C_k$. This is exactly the factorisation asserted in Cor. 9.13. □

MD alignment (micro-note). "Guided" stewards are the stewardship loci where value-alignment is ensured (either by a Stage–D agent or by a Stage–D teacher). The factorisation says: *any* guidance assignment that respects stewardship and alignment *must* land inside Guided_k; this makes the governance layer computably crisp and modal-stable.

H Instance Fibration and Election Dynamics

Let **Lev** be the thin category on the chain $1 \to 2 \to \cdots \to 10$. Write $i_{k,\ell} : C_k \hookrightarrow C_\ell$ for the levelwise council inclusions from §9 (so $i_{k,k} = \mathrm{id}$ and $i_{k,m}\, i_{m,\ell} = i_{k,\ell}$).

Foundational stance. All arguments are internal to the ambient topos and use only finite limits, disjoint finite coproducts, images, and left-exactness of the five value nuclei.

Definition of the instance category

For each level k, let Inst_k be the full subcategory of the slice $\mathscr{E}/C_k$ on *ledger–clopen* arrows $\alpha : X \to C_k$ with $X \subseteq \mathrm{W}_k$ (typed to level k). A morphism in Inst_k is a commuting triangle $X \xrightarrow{h} Y \xrightarrow{\beta} C_k$ with $\beta \circ h = \alpha$. Define the category of instances Inst as the Grothendieck construction of the pseudo–functor

$$\mathbf{Lev} \longrightarrow \mathbf{Cat}, \qquad k \longmapsto \mathsf{Inst}_k,$$

with total objects the pairs $(k, \alpha : X \to C_k)$ and arrows

$$(k, \alpha) \xrightarrow{(s,h)} (\ell, \beta) \quad \text{over} \quad s : k \to \ell$$

given by a map $h : X \to \mathrm{dom}(\beta)$ with $\beta \circ h = i_{k,\ell} \circ \alpha$. The projection $p : \mathsf{Inst} \to \mathbf{Lev}$ is $(k, \alpha) \mapsto k$.

Guidance/eligibility side–conditions. Let $\mathsf{Steward}_k \hookrightarrow C_k$ and $\mathsf{Guided}_k \hookrightarrow \mathsf{Steward}_k$ be the clopen subobjects from §9 (see also Appendix G). Fix:

- an *eligibility* predicate Elig_k on C_k (Def. 9.10);
- a *guidance* predicate Align_k on C_k (Constraint 9.11), with $\mathsf{Guided}_k = \mathsf{Steward}_k \wedge \mathrm{Align}_k$;
- *compatibility along inclusions*: $i^*_{k,\ell}(\mathrm{Elig}_\ell) \Rightarrow \mathrm{Elig}_k$ and $i^*_{k,\ell}(\mathrm{Align}_\ell) \Rightarrow \mathrm{Align}_k$.

We restrict Inst_k to those $\alpha : X \to C_k$ whose graph lies inside $\mathsf{Steward}_k \wedge \mathrm{Elig}_k$ (so α already "picks" eligible stewards).

Direct image along levels (postcomposition)

For $s : k \to k{+}1$ in $\mathbf{Lev}$, define the *direct image*

$$s_! \;:=\; (i_{k,k+1})_! \;:\; \mathsf{Inst}_k \;\longrightarrow\; \mathsf{Inst}_{k+1}, \qquad s_!(\alpha) \;:=\; i_{k,k+1} \circ \alpha,$$

and on arrows $h : (k, \alpha) \to (k, \beta)$ put $s_!(h) := h$. This is well–defined: domains are unchanged (hence ledger–clopen), and by compatibility $i_{k,k+1} \circ \alpha$ remains inside $\mathsf{Steward}_{k+1} \wedge \mathrm{Elig}_{k+1}$. For a composite $k \to \ell$ put $F_{k\to\ell} := (i_{k,\ell})_!$.

Lemma H.1 (Canonical lift over a successor)**.** *For each (k, α) there is an arrow*

$$\chi_\alpha : \; (k, \alpha) \;\longrightarrow\; (k{+}1,\, i_{k,k+1} \circ \alpha)$$

over $k \to k{+}1$ whose component on domains is $\mathrm{id}_{\mathrm{dom}(\alpha)}$.

Proof. Immediate from the definition of arrows in Inst: the square $\mathrm{dom}(\alpha) \xrightarrow{\mathrm{id}} \mathrm{dom}(\alpha)$ with codomain map $i_{k,k+1} \circ \alpha$ commutes. □

Opcartesianness and cleavage

Proposition H.2. *The projection $p : \mathsf{Inst} \to \mathbf{Lev}$ is a cloven opfibration. For each (k, α) the chosen opcartesian lift of $k \to k{+}1$ is χ_α from Lemma H.1. Consequently, for any $k \le \ell$, the arrow $(k, \alpha) \to (\ell,\, i_{k,\ell} \circ \alpha)$ with identity on the domain is opcartesian over $k \to \ell$.*

Proof. Fix $s : k \to k{+}1$ and χ_α as above. We verify the opcartesian universal property. Suppose we are given $t : k{+}1 \to \ell$ and an arrow

$$(k, \alpha) \xrightarrow{(t\circ s,\, h)} (\ell, \gamma), \qquad \text{so} \qquad \gamma \circ h \;=\; i_{k,\ell} \circ \alpha.$$

Define $u := (t,\, h) : \; (k{+}1,\, i_{k,k+1} \circ \alpha) \to (\ell, \gamma)$. This is a valid arrow because $\gamma \circ h = i_{k,\ell} \circ \alpha = i_{k+1,\ell} \circ (i_{k,k+1} \circ \alpha)$. Then $u \circ \chi_\alpha = (t{\circ}s,\, h)$. Uniqueness: if $u' : (k{+}1,\, i_{k,k+1} \circ \alpha) \to (\ell, \gamma)$ satisfies $u' \circ \chi_\alpha = (t{\circ}s,\, h)$, its domain component must be h (since χ_α is identity on domains), and the base component must be t. Hence $u' = u$. Thus χ_α is opcartesian. The same argument, iterated, yields the claim for $k \to \ell$. Choosing χ_α for each (k, α) gives a cleavage. □

Remark (bifibration viewpoint). The construction is the pullback of the standard codomain bifibration $\mathrm{cod} : \mathrm{Arr}(\mathscr{E}) \to \mathscr{E}$ along the functor $k \mapsto C_k$. Opcartesian lifts are just postcomposition with $i_{k,\ell}$, cartesian lifts (not used here) would be pullbacks along $i_{k,\ell}$.

Preservation of ledger–clopens and guidance

Lemma H.3 (Stability under level reindexing)**.** *For any arrow $s : k \to \ell$ in* **Lev***:*

1. *The direct image $s_!$ preserves ledger–clopen instances.*
2. *If α satisfies the eligibility and stewardship side–conditions ($\Gamma_\alpha \subseteq \mathsf{Steward}_k \wedge \mathrm{Elig}_k$), then $s_!(\alpha)$ satisfies the corresponding conditions at level ℓ.*
3. *If, in addition, α factors through Guided_k, then $s_!(\alpha)$ factors through $i_{k,\ell}(\mathsf{Guided}_k) \subseteq C_\ell$; and when Guided_ℓ is* levelwise compatible *with inclusions ($i_{k,\ell}(\mathsf{Guided}_k) \le \mathsf{Guided}_\ell$; this is part of Constraint 9.11), then $s_!(\alpha)$ lands in Guided_ℓ.*

Proof. (1) Domains are unchanged by postcomposition, so ledger–clopens are preserved. (2) By compatibility $i^*_{k,\ell}(\mathrm{Elig}_\ell) \Rightarrow \mathrm{Elig}_k$, hence $\Gamma_{s_!(\alpha)} \subseteq \mathsf{Steward}_\ell \wedge \mathrm{Elig}_\ell$. (3) Immediate from factoring α through Guided_k and the assumed monotonicity $i_{k,\ell}(\mathsf{Guided}_k) \le \mathsf{Guided}_\ell$. $\square$

Election functor and nomination arrow. For each successor $k \to k{+}1$, define

$$F_{k\to k+1}(k,\alpha) \;:=\; (k{+}1,\, i_{k,k+1} \circ \alpha), \qquad \eta_\alpha \;:=\; \chi_\alpha :\; (k,\alpha) \longrightarrow F_{k\to k+1}(k,\alpha).$$

By Proposition H.2, η_α is opcartesian (*nomination/election* arrow). By Lemma H.3, $F_{k\to k+1}$ preserves ledger–clopens and the guidance constraints.

MD alignment (micro–note). An instance is "what a council sees" at a level; moving up a level is post-composition along the council ladder. Opcartesianness says that any higher-level assignment from a lower-level instance factors uniquely through the canonical nomination/election arrow. This makes election dynamics *computably crisp* and nucleus-stable.

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Programme note. This manuscript is part of a broader effort to lay precise categorical foundations for a new scientific theory—"MD Theory"—corresponding to Madhyasth Darshan. The present paper supplies the decidable 1-topos core; companion works develop the modal calculus and SK–HCK–CK bridge (Paper 0, Paper A) and the higher-topos upgrade (Paper B) [22, 23, 24].

Societal motivation. The backdrop of war and climate instability underscores a long-felt need: a *consistent and correct* account of the human dimension that can unite purpose and cooperation. Although the MD framework has been available since 1972 [18, 19], scholars have often hesitated to connect its philosophy with formal science and mathematics. This project aims to build that bridge—showing how MD's commitments can be rendered precise, testable, and computable—so that shared understanding can guide humane systems in education, governance, and economics.

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License and Ethical Rationale

 In Madhyasth Darshan (MD), the very substance of knowledge is identified with *Brahma*, formalised here as the strict terminal object $\mathbf{1}_B$ (philosophically denoted "THIS"). As a consequence, knowledge is treated as an all-pervading public good. Enclosing it for commercial gain—or modifying it to evade open, cooperative use—would obstruct the 10–Tier World–Family programme in HCK and risk unjust social relations.

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MD alignment (micro-note). Making the ledger and its proofs openly available aligns the formal programme with MD's view that value-knowledge flows from $\mathbf{1}_B$ and should remain universally accessible—supporting education, governance, and humane systems within the CK $\subset$ HCK $\subset$ SK bridge.

Conflict of Interest

The author declares no conflict of interest.

[4]See creativecommons.org/licenses/by-nc-sa/4.0/.

References

[1] P. T. Johnstone, *Sketches of an Elephant: A Topos Theory Compendium*, 2 vols., Oxford Logic Guides **43–44**, Oxford University Press, 2002.

[2] A. Joyal and M. Tierney, *An Extension of the Galois Theory of Grothendieck*, Memoirs of the American Mathematical Society **51** (No. 309), 1984. DOI: `10.1090/memo/0309`.

[3] I. Moerdijk and E. Palmgren, *Type theories, toposes and constructive set theory: Predicative aspects of AST*, Annals of Pure and Applied Logic **114** (2002), 155–201.

[4] S. Mac Lane and I. Moerdijk, *Sheaves in Geometry and Logic: A First Introduction to Topos Theory*, Universitext, Springer, 1992.

[5] O. Caramello, *Theories, Sites, Toposes*, Oxford University Press, 2018.

[6] P. Deligne (ed.), *Cohomologie Étale: SGA* $4\frac{1}{2}$, Lecture Notes in Mathematics **569**, Springer, 1977.

[7] P. T. Johnstone, *Factorization theorems for geometric morphisms*, Cahiers de Topologie et Géométrie Différentielle Catégoriques **22** (1981), no. 1, 3–16.

[8] P. T. Johnstone and I. Moerdijk, *Local maps of toposes*, Proceedings of the London Mathematical Society (3) **58** (1989), 281–305.

[9] S. Vickers, *Topology via Logic*, Cambridge Tracts in Theoretical Computer Science **5**, Cambridge University Press, 1989.

[10] M. Makkai and G. E. Reyes, *First Order Categorical Logic: Model-Theoretical Methods in the Theory of Topoi and Related Categories*, Lecture Notes in Mathematics **611**, Springer, 1977.

[11] C. C. Chang and H. J. Keisler, *Model Theory*, 3rd ed., Studies in Logic and the Foundations of Mathematics **73**, North-Holland, 1990.

[12] A. Joyal and I. Moerdijk, *Algebraic Set Theory*, London Mathematical Society Lecture Note Series **220**, Cambridge University Press, 1995.

[13] J. Lurie, *Higher Topos Theory*, Annals of Mathematics Studies **170**, Princeton University Press, 2009. DOI: `10.1515/9781400830558`.

[14] P. Scholze, *Lectures on Condensed Mathematics* (lecture notes), University of Bonn, 2019. Available at https://www.math.uni-bonn.de/people/scholze/Condensed.pdf.

[15] J. M. E. Hyland, P. T. Johnstone, and A. M. Pitts, *Tripos theory*, Mathematical Proceedings of the Cambridge Philosophical Society **88** (1980), 205–232.

[16] D. Clausen and P. Scholze, *Condensed Mathematics and Complex Geometry*, 2022. Available at https://people.mpim-bonn.mpg.de/scholze/Complex.pdf.

[17] B. Bhatt and P. Scholze, *The pro-étale topology for schemes*, 2013. Preprint, https://arxiv.org/abs/1309.1198.

[18] A. Nagraj. *Manav Vyavhār Darshan.* Amarkantak: Jeevan Vidya Prakashan, 1972. [First MD text; in Hindi.] Online: https://www.madhyasth-darshan.info.

[19] A. Nagraj. *Manav Vyavhār Darshan*, 2011 ed.; reprinted 2019. Amarkantak: Jeevan Vidya Prakashan; copyright © Divyapath Sansthān, Amarkantak. [Foundational MD text; in Hindi.] Online: https://www.madhyasth-darshan.info.

[20] B. Meena. *From the SSS Principle to the Riemann Hypothesis and Beyond: A Unified Higher–Topos Framework.* Zenodo, 2025. Version 2. DOI: 10.5281/zenodo.16784277. Latest version (concept DOI): 10.5281/zenodo.15811994.

[21] B. Meena. *The Ātmā-Atom as a 5-Leg Local Shtuka and a Solution of the Kottwitz Conjecture for Its Moduli.* Zenodo, 2025. Version 2. DOI: 10.5281/zenodo.16784291. Latest version (concept DOI): 10.5281/zenodo.15814811. [SU(5) 61–dim split; $1, 2, 8, 18, 32$ blocks; projector realisation for MD.]

[22] B. Meena. *P0: Idempotent Monads, Boolean Phase and Collective Atoms in the Madhyasth Darshan Topos.* Zenodo, 2025. Version 3. DOI: 10.5281/zenodo.16784305. Latest version (concept DOI): 10.5281/zenodo.16269489.

[23] B. Meena. *Paper A: Modal Calculus and Monadic Framing for MD.* Zenodo, 2025. Version 1. DOI: 10.5281/zenodo.16784270.

[24] B. Meena. Paper B: Modal–Homotopy Shape and SK Completeness in the MD–Topos. Preprint, 2025. (Companion to [22, 23].)